



General Physics II

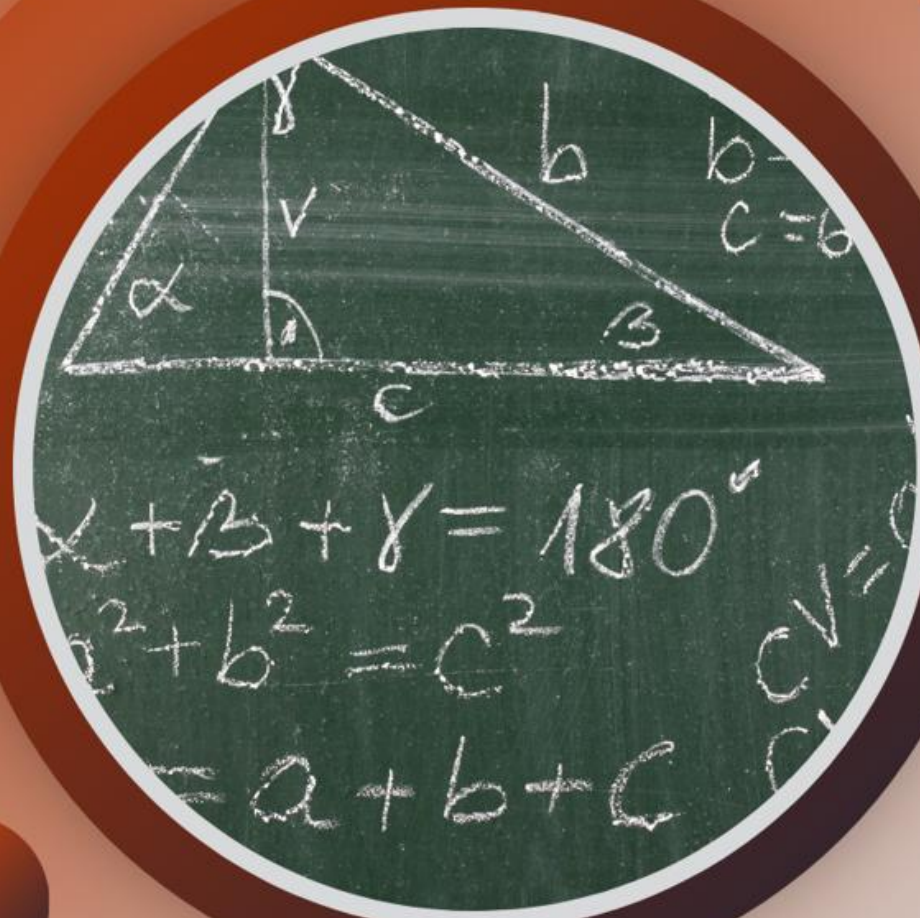
PHY121

College of Science and Technology



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MODULE 1: ELECTROSTATICS

UNIT 1: Forces in Nature & Introduction to Electrostatics

LEARNING OUTCOMES

At the end of this unit, you should be able to explain:

- Identify the four fundamental forces in nature
- explain the concept of electric charge and its quantisation
- describe charging by friction, conduction and induction.

INTRODUCTION

- Everything around us, from the way a balloon sticks to a wall after rubbing, to lightning flashing across the sky, is governed by a few basic forces in nature.
- These forces are the “rules” that explain how objects in the universe interact with each other. Scientists have identified four fundamental forces: gravity, the electromagnetic force, the strong nuclear force, and the weak nuclear force.
- Among these, the electromagnetic force is the one we encounter most often in daily life, because it is responsible for electricity, magnetism, and even light.



Gravitational Force
Acts between masses



Electromagnetic Force
Acts between charges



Strong Nuclear Force
Holds the nucleus together



Weak Nuclear Force
Responsible for
radioactive decay

- **Electrostatics is a branch of physics that focuses on electric charges at rest. In simple terms, it studies what happens when electric charges are at rest and interact with each other.**
- **Even though the charges are not moving, they can still exert a force on each other through space. This invisible interaction is what we call the electric force.**
- **At the heart of electrostatics is the idea of electric charge. Charge is a basic property of matter, and it comes in two types: positive and negative. Like charges repel, while opposite charges attract.**
- **One important fact about charge is that it cannot disappear or be created from nothing; it can only move from one object to another. Another key idea is that charge exists in small fixed amounts, rather than any random value, which means it is “quantized.”**

- **Electrostatics also helps us understand how materials behave. Conductors allow charges to move freely, while insulators hold charges in place.**
- **This difference is very important in designing electrical devices, protecting sensitive equipment, and even in technologies like capacitors that store electrical energy.**
- **Beyond everyday experiences, electrostatics plays a major role in modern science and technology.**
- **It is used in devices such as photocopiers, air purifiers, medical imaging tools, and many electronic systems. It also forms the foundation for more advanced topics in physics, including electric circuits, magnetism, and electromagnetic waves.**

- In summary, studying forces in nature and electrostatics helps us understand the invisible forces that shape much of the physical world.
- It connects simple everyday experiences to powerful scientific ideas and lays the groundwork for understanding how electricity and technology work in modern life.



Rubbing a comb attracts paper bits.



A charged balloon sticks to the wall.



Lightning occurs due to charge separation in clouds.



Photocopiers use electrostatic charges to form images.



Electrostatic precipitators remove dust and smoke from air.



Capacitors store energy in electric fields.

ELECTRIC CHARGES and its properties

- **Electric charge is a fundamental property of matter responsible for electrical interactions.**
- **It is the basis of electricity, magnetism, and electrostatics.**
- **Electric charge is the property of matter that causes it to experience force in an electric field.**
- **Symbol: Q or q**
- **Unit: Coulomb (C)**

Types of electric charge

- 1. Positive charge (+): carried by protons
- 2. Negative charge (–): carried by electrons
- Like charges repel, unlike charges attract.

Interaction of charges

- Charges interact through electrostatic force.
- Like charges repel; unlike charges attract.

Properties of charges

1. additivity of charge

- Total charge is the algebraic sum of individual charges:
- $Q_{\text{total}} = q_1 + q_2 + q_3 + \dots$

2. conservation of charge

- Charge cannot be created or destroyed, only transferred.
- Total charge in an isolated system remains constant.

3. Quantization of Charge

- Charge exists in discrete units:
- $q = ne$, where $e = 1.602 \times 10^{-19} \text{ C}$

$$q = ne$$

Where:

- $n = \text{integer } (0, \pm 1, \pm 2, \pm 3, \dots)$
 - $e = 1.602 \times 10^{-19} \text{ C}$
- **4. Invariance of Charge**
 - The magnitude of charge does not depend on:
 - i. Speed of the object
 - ii. Reference frame
 - iii. Physical conditions (temperature, pressure)
 - A charge remains the same whether it is moving or at rest.

- **5. Scalar Nature of Charge**

- Electric charge is a scalar quantity:
- It has a magnitude only
- It does not have a direction
- Signs (+/–) indicate type, not direction

- **6. Transferability of Charge**

- Charge can be transferred between bodies through:
- Conduction (direct contact)
- Induction (without contact)
- Friction (rubbing)

Methods of Charging Bodies

(a) Charging by Friction

- Rubbing two materials transfers electrons

(b) Charging by Conduction

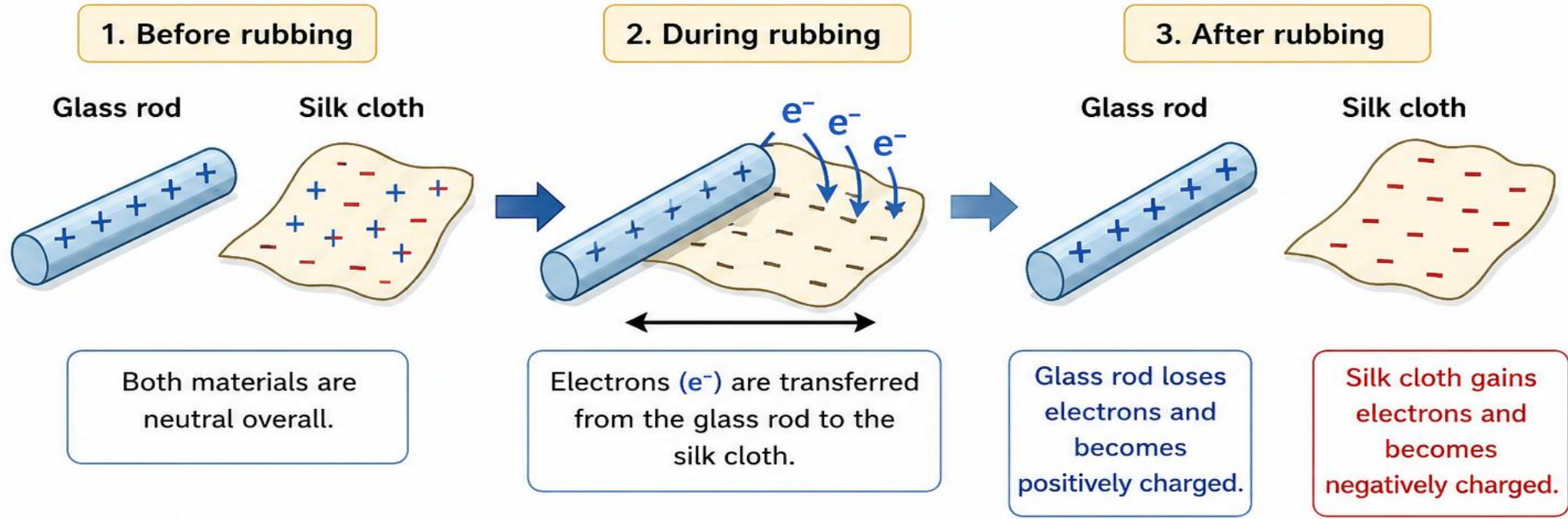
- Direct contact between charged and neutral objects

(c) Charging by Induction

- Redistribution of charges without contact

(a) CHARGING BY FRICTION

Rubbing two materials transfers electrons.



Key:



Positive charge
(deficit of electrons)



Negative charge
(excess of electrons)



Electron

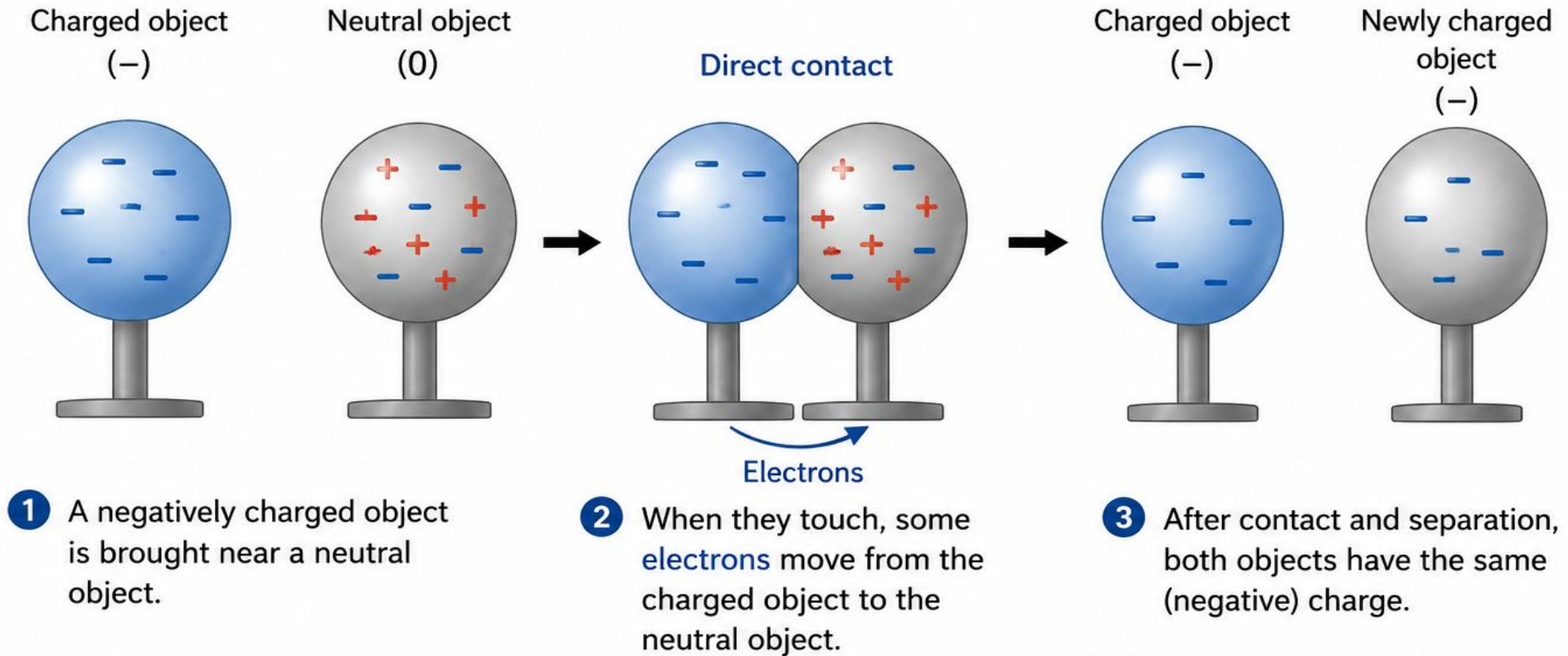


Direction of electron
transfer

Charging by friction involves transfer of electrons. One body loses electrons and becomes positively charged, while the other gains electrons and becomes negatively charged.

(b) Charging by Conduction

Direct contact between charged and neutral objects.

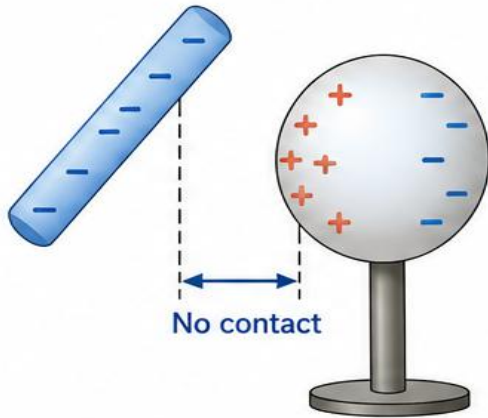


Key idea: Charge is transferred by the flow of electrons during direct contact.

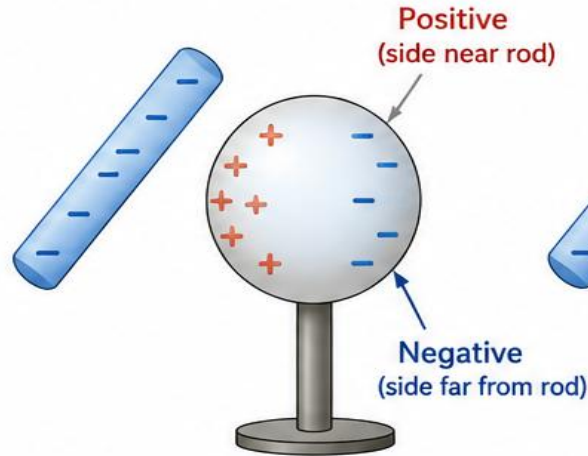
Charging by Induction

Redistribution of charges without contact.

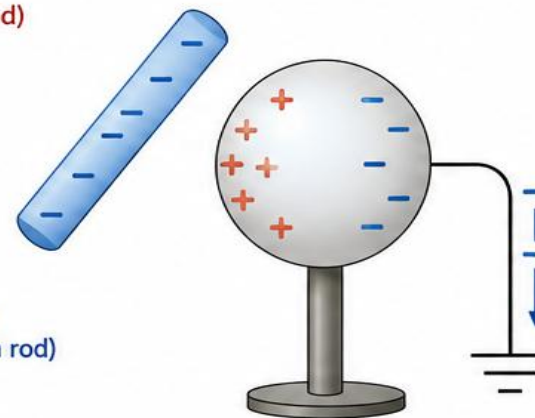
- 1 A negatively charged object is brought near a neutral conductor without touching it.



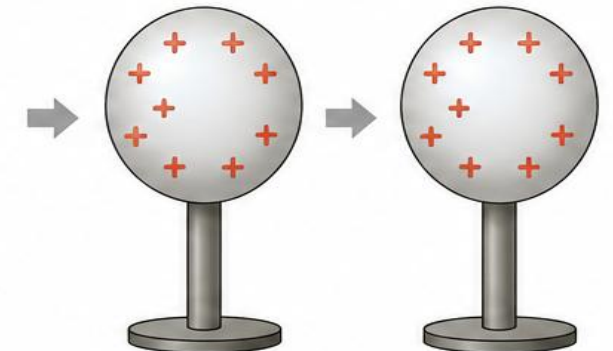
- 2 Charges in the conductor redistribute: electrons are repelled to the far side, leaving positive charges on the near side.



- 3 While the rod is still near, the conductor is earthed. Electrons flow to ground (away from the conductor).



- 4 The earth connection is removed first, then the rod is removed. The conductor is left with a net positive charge.



+ Positive charge
 - Negative charge

Key idea: A nearby charged object causes charge redistribution in a conductor. If earthed while polarized, some charges leave or enter, leaving the conductor charged.



MODULE 1: ELECTROSTATICS

UNIT 2: Coulomb's Law and Electric Field Concept

INTRODUCTION

- **Electrostatics deals with electric charges at rest and the forces between them. Two important concepts in electrostatics are:**
- **Coulomb's Law — explains the force between electric charges.**
- **Electric Field — explains how charges influence the space around them.**
- **These concepts form the foundation of electricity, electronics, electromagnetism, and many modern technologies.**

LEARNING OUTCOMES

At the end of this unit, you should be able to :

- State and apply coulomb's law
- calculate forces between multiple charges using superposition
- define and interpret electric field lines

**opposite charges attract
because of the
electric force**

Coulomb's Law

$$\mathbf{F} = k \frac{q_1 q_2}{r^2}$$

**magnitude of
electric force**

**opposite charges attract
because of the
electric force**

Coulomb's Law

$$\mathbf{F} = \mathbf{k} \frac{\mathbf{q}_1 \mathbf{q}_2}{\mathbf{r}^2}$$

**Coulomb's constant
($8.99 \times 10^9 \text{ N m}^2/\text{C}^2$)**

electric force can be +/-
(q can be +/-)

gravitational force must be +
(m must be +)

Coulomb's Law

$$F = k \frac{q_1 q_2}{r^2}$$

Newton's Law of Universal Gravitation

$$F = G \frac{m_1 m_2}{r^2}$$

Coulomb's constant: 8.99×10^9

gravitational constant: 6.67×10^{-11}

almost a billion trillion times larger

Coulomb's Law

$$F = k \frac{q_1 q_2}{r^2}$$

Newton's Law of Universal Gravitation

$$F = G \frac{m_1 m_2}{r^2}$$

electric force **increases**
as **charge** (q) increases

Coulomb's Law

$$F = k \frac{q_1 q_2}{r^2}$$

electric force **decreases**
as **distance** (r) increases

(more than two charges
require **vector** analysis)

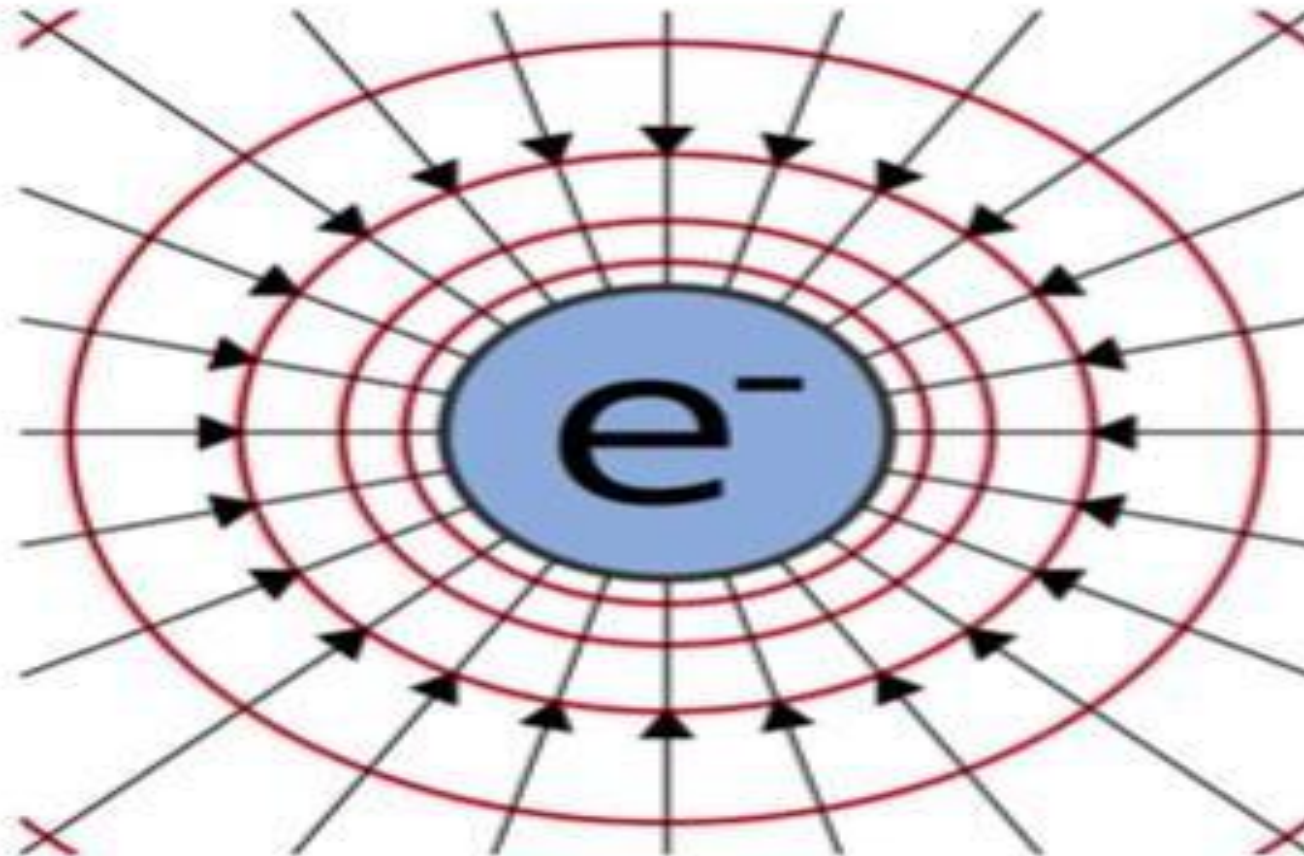


the gravitational **force** arises
from a gravitational **field**

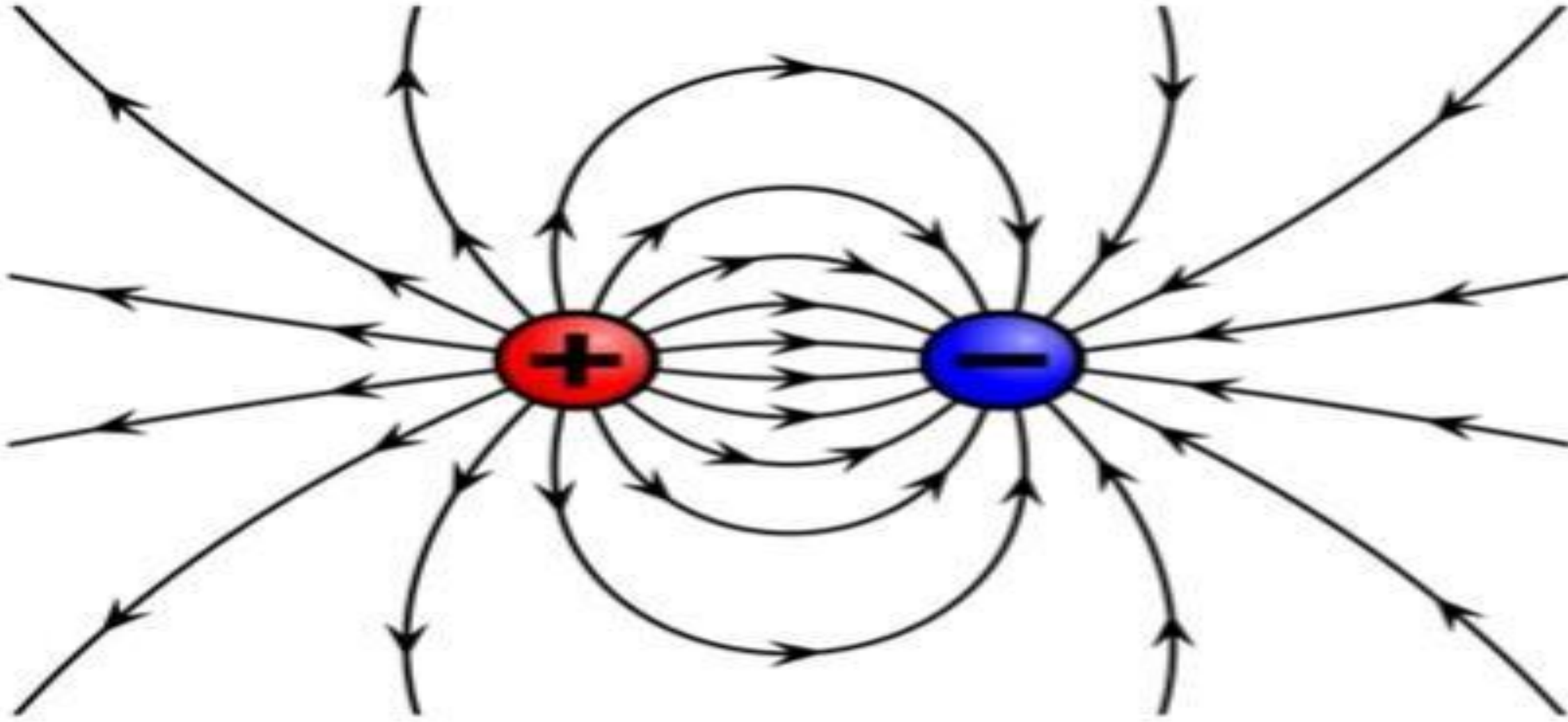
the electric **force** arises
from a electric **field**

$$F_E \gg F_G$$

**any charged object
creates an electric field**



these fields interact

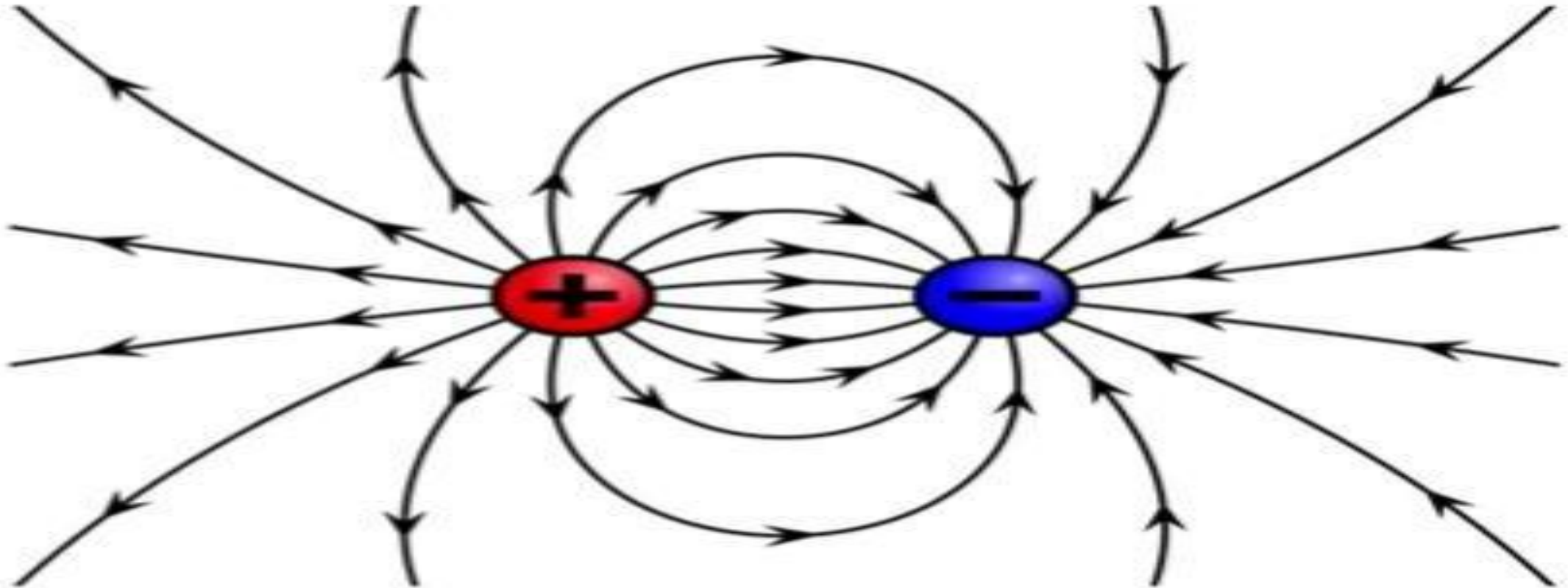


electric field strength

$$E = k_c \frac{q}{r^2}$$

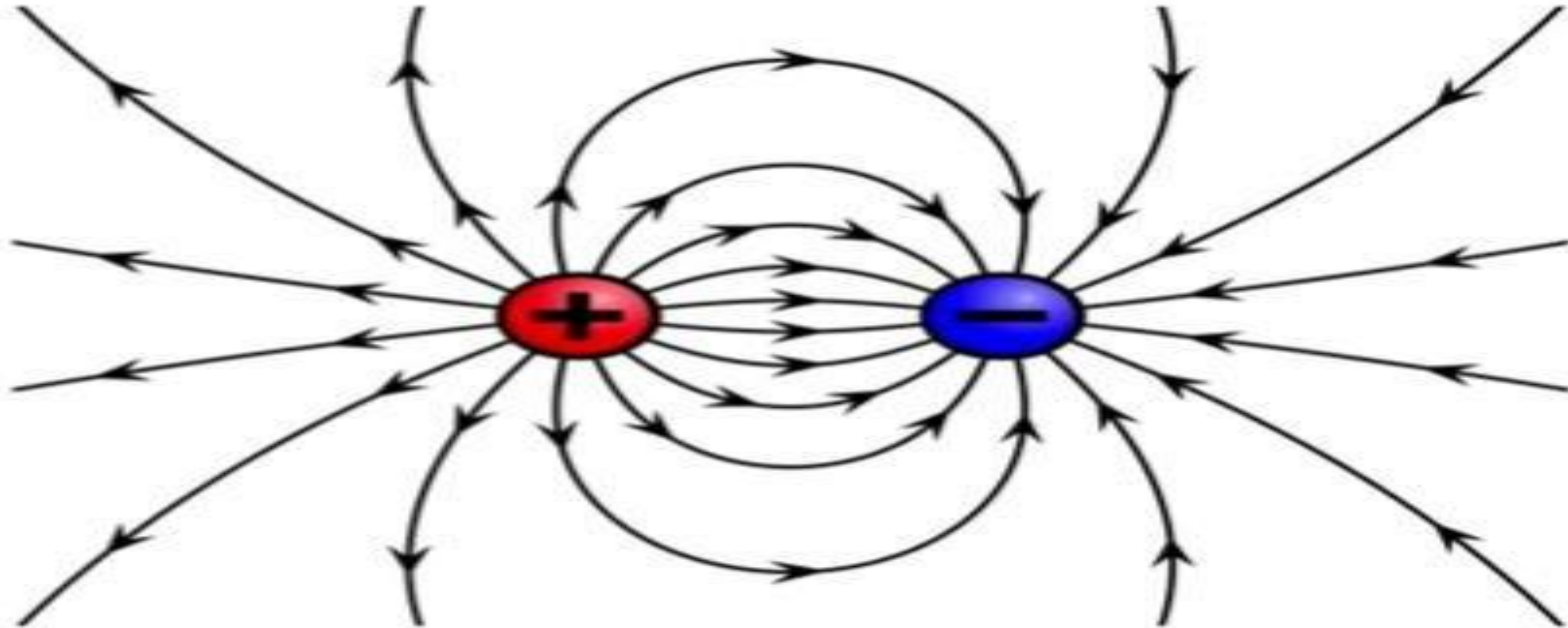
Coulomb's constant

electric field lines



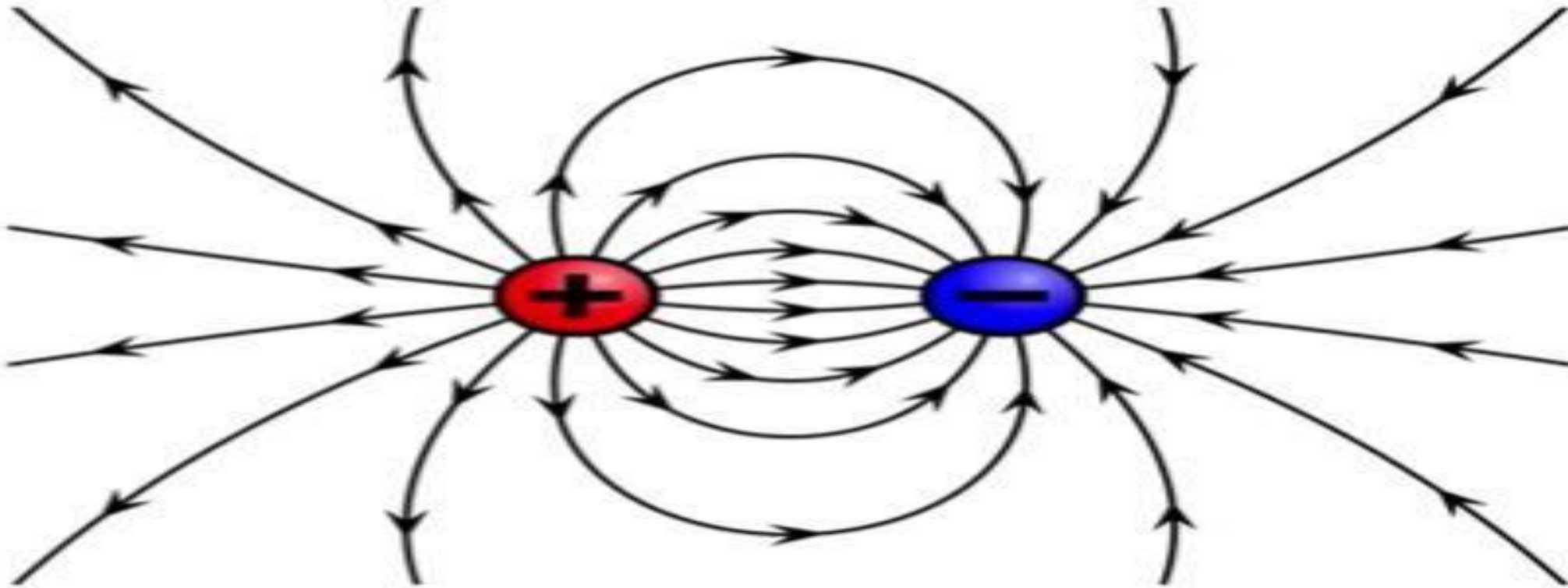
point **towards** negative charges
point **away from** positive charges

electric field lines



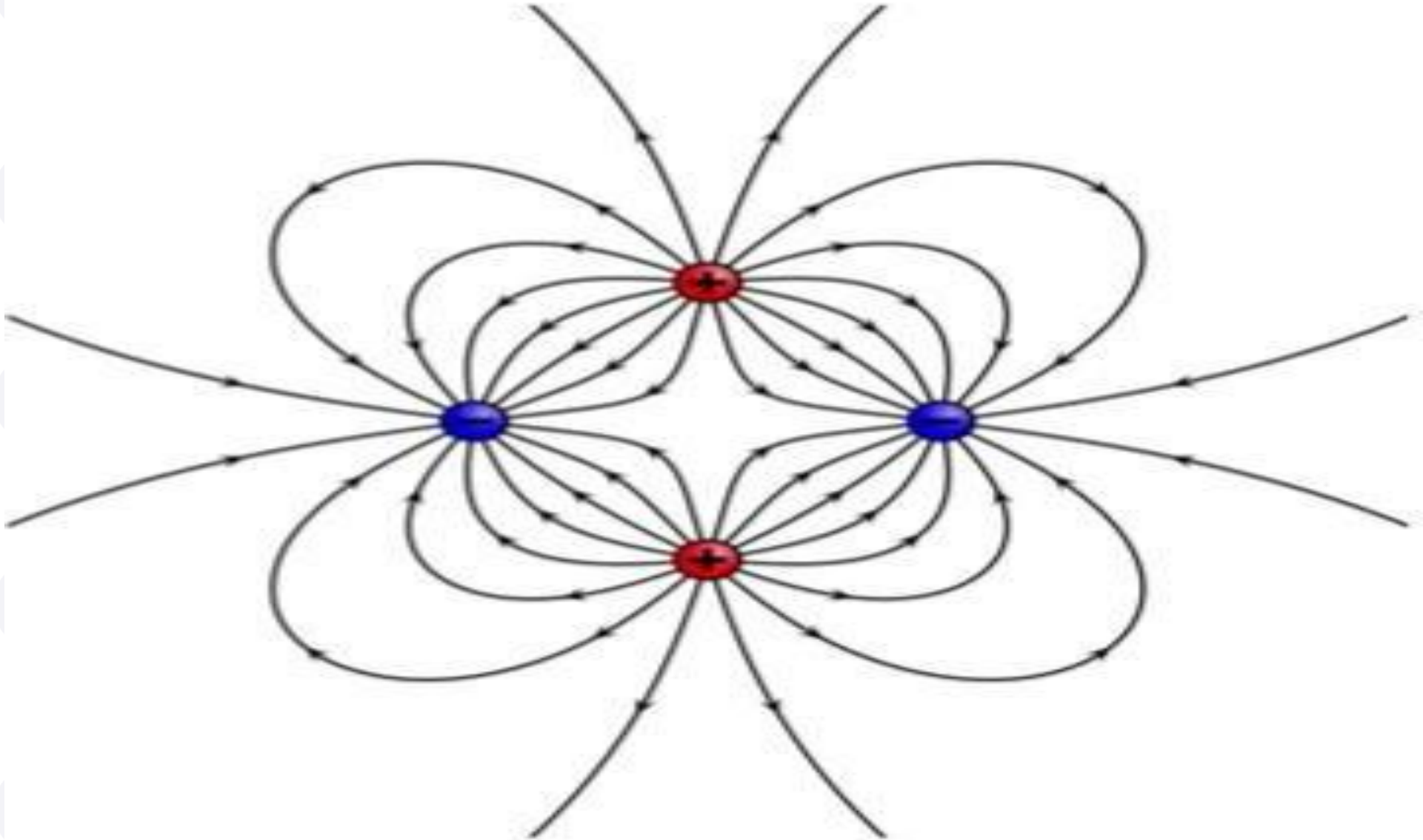
electric dipole

electric field lines



more lines = stronger field

electric field lines



**What is the magnitude of the electric force
between two protons five meters apart?**

What is the magnitude of the electric force between two protons five meters apart?

$$F = k_c \frac{q_1 q_2}{r^2}$$

$$F = 8.99 \times 10^9 \text{ N m}^2/\text{C}^2 \frac{(1.6 \times 10^{-19} \text{ C})^2}{(5.0 \text{ m})^2}$$

$$F = 9.2 \times 10^{-30} \text{ N}$$

**What do we
do with the
electric force?**

**let's learn about
electric
potential**

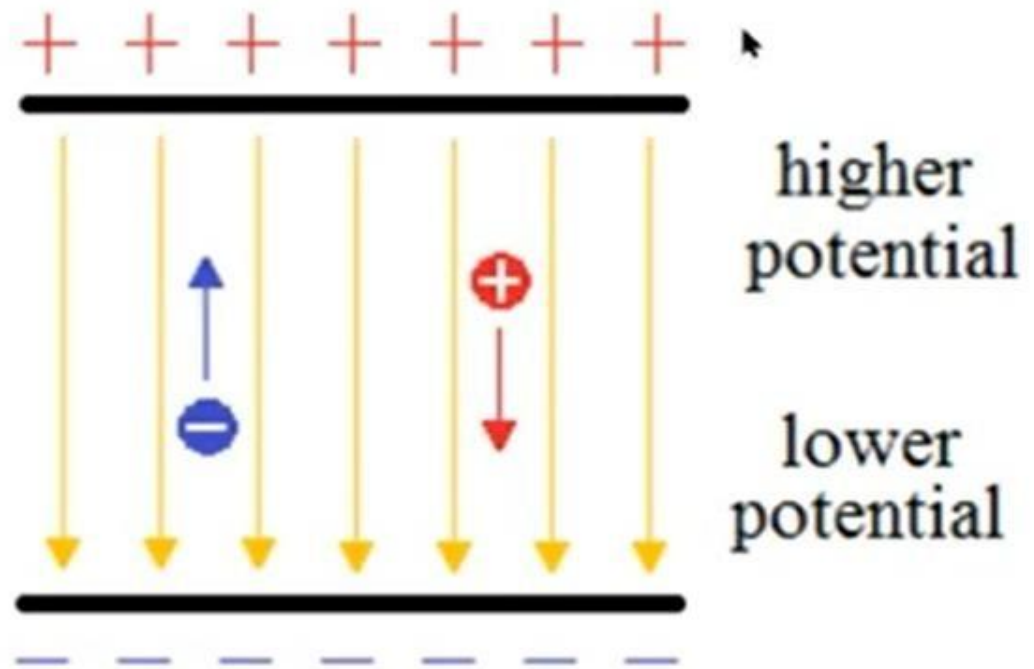
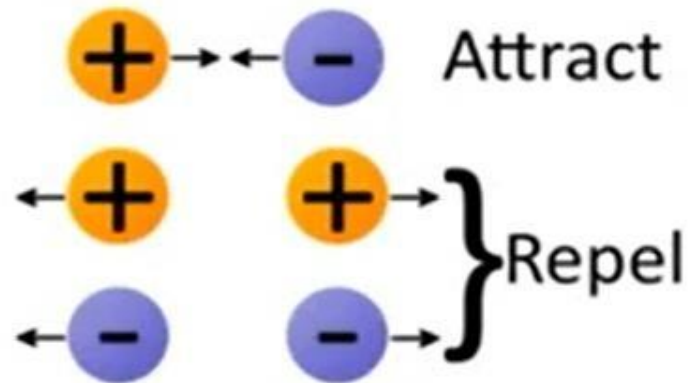
Electric Fields

When a ferromagnetic object is placed close to a magnet, it will experience an attractive force – the magnetic field

Similarly, a charged object has an electric field

An electric field is a region of space in which a charged object will experience an electrostatic force

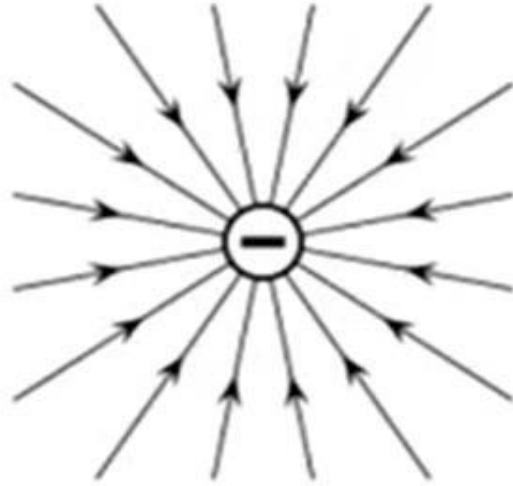
The direction of the electric field line at a point is the direction in which a **positive** charge point would accelerate, when placed at that point



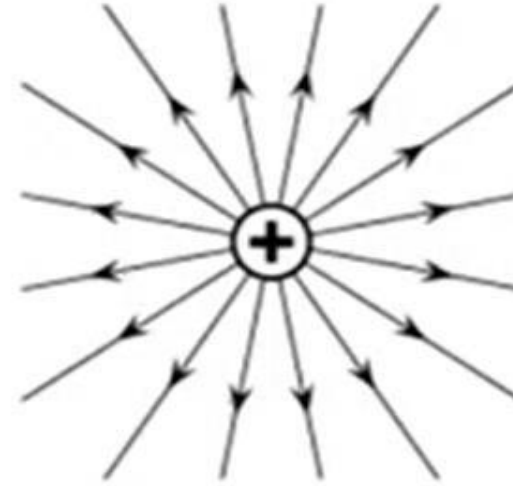
Field lines:

- Are imaginary lines that represent an electric field
- Are in the direction that a positive point charge would move if placed in a field
- Never cross or touch each other
- Start and end perpendicular to the surface of the charged object
- Exist in three dimensions around a charged point or object
- Indicate a strong electric field when close together (high density)
- Indicate a weak field when far apart (low density)

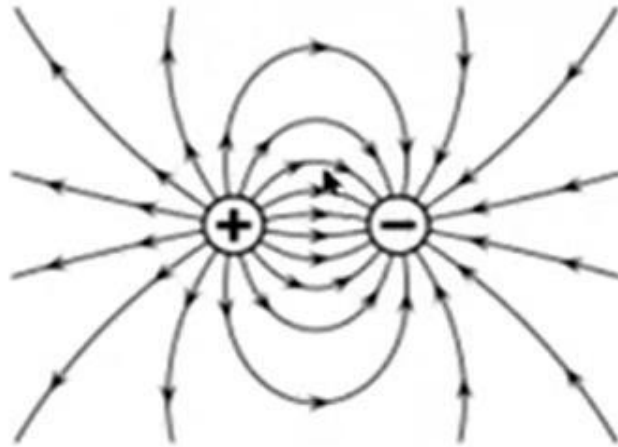
Around a negative point charge



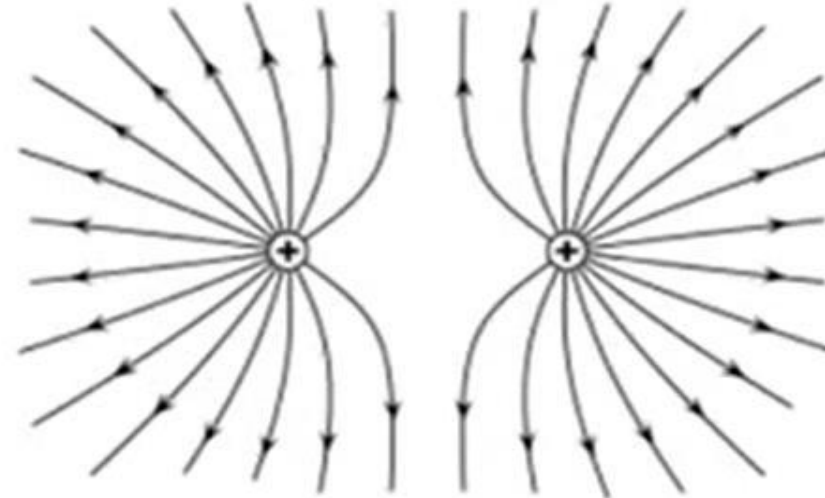
Around a positive point charge



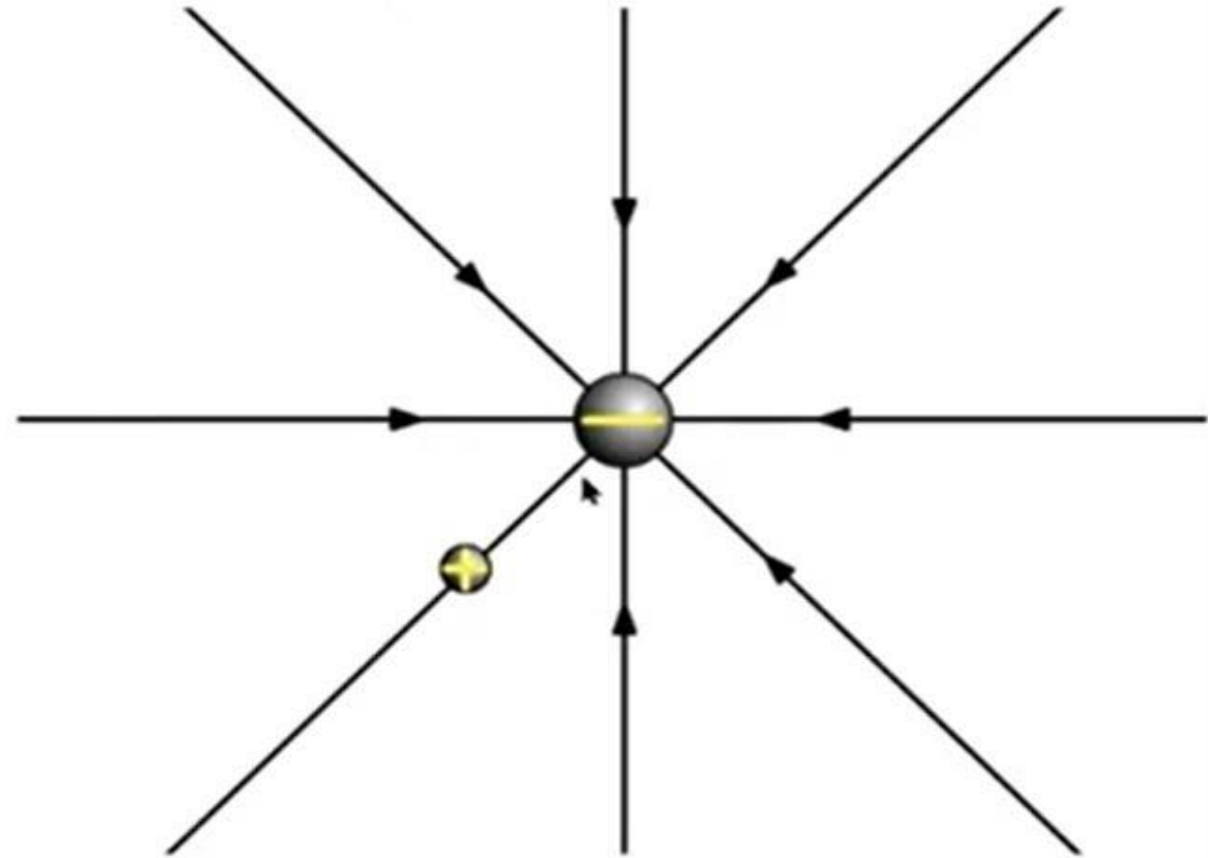
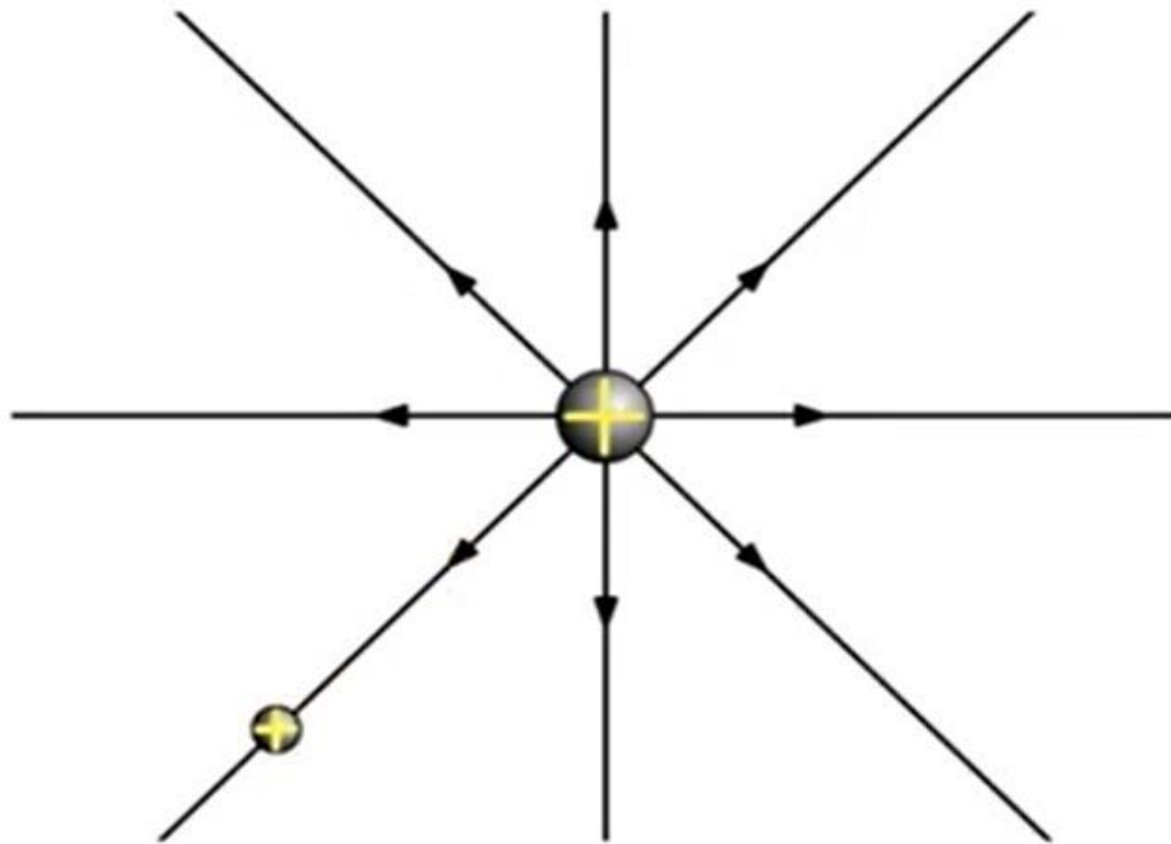
Between unlike point charges



Between like point charges



The direction of the electric field at a point is the direction that a positive test charge would move if placed at that point



What is the direction of the field 1 m away from the 1C charge, in or out of the 1C charge?

The electric field strength at a point: The electric field strength at a point is the electrostatic force experienced per unit positive charge placed at that point.

Electric field
($\text{N}\cdot\text{C}^{-1}$)

Force (N)

$$E = \frac{F}{q}$$

or

$$F = qE$$

Charge(C) on which the electrostatic force is exerted

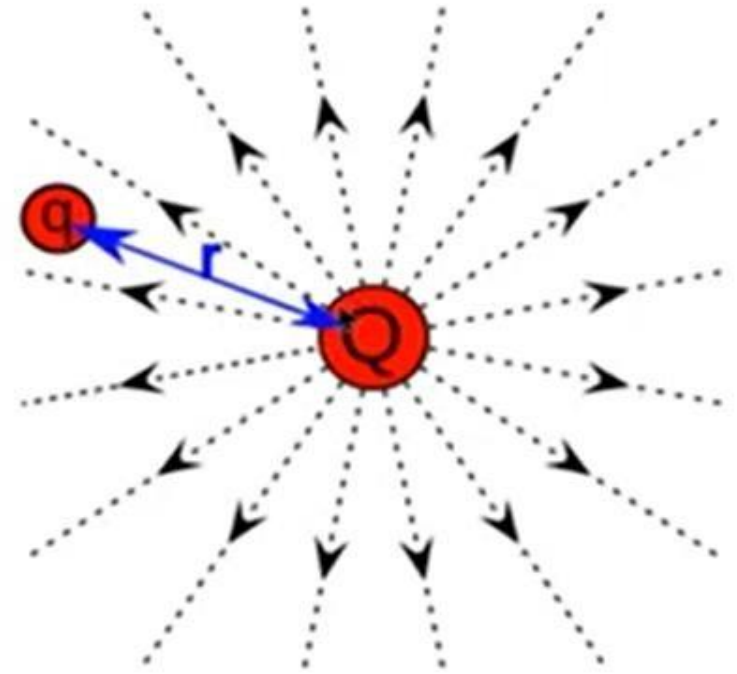
Note that:
 $1 \text{ N}\cdot\text{C}^{-1} = 1 \text{ V}\cdot\text{m}^{-1}$

If a charge q is placed in an electric field produced by charge Q , the force on q is F and the electric field caused by Q is E , and the distance between the two charges is r

$$F = \frac{kQq}{r^2} \quad \text{and} \quad F = qE$$

therefore

$$\cancel{qE} = \frac{\cancel{kQq}}{r^2} \quad \text{so} \quad E = \frac{kQ}{r^2}$$



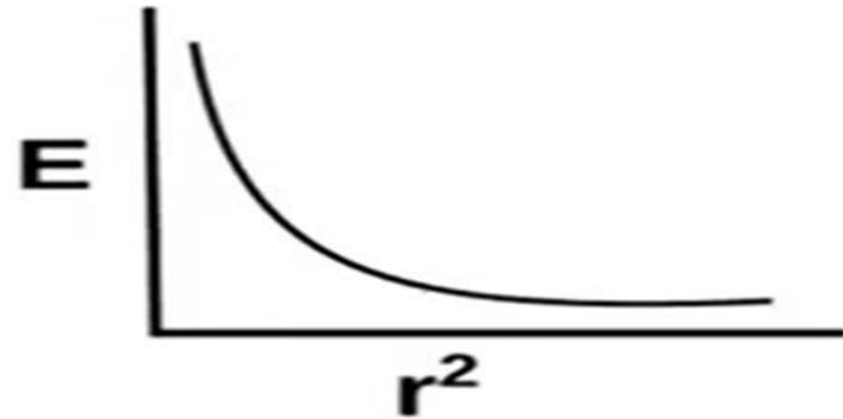
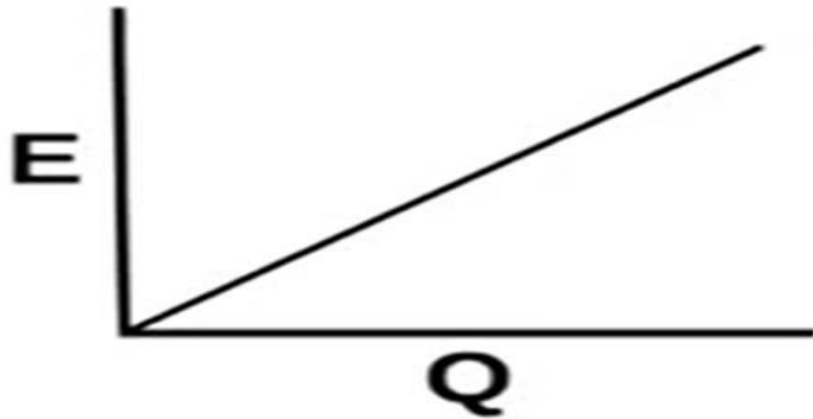
$E \propto Q$ – electric field is directly proportional to the magnitude of the charge causing the field

And

Electric field
($\text{N}\cdot\text{C}^{-1}$)

$$E \propto \frac{1}{r^2}$$

Distance between
point charges (m)



Examples :

A charge, Q , causes an electric field, E at a distance r . By which factor will the field strength E change if the charge changes to $5Q$ and the distance changes to $3r$?

Examples :

A charge, Q , causes an electric field, E at a distance r . By which factor will the field strength E change if the charge changes to $5Q$ and the distance changes to $3r$?

$$E \propto \frac{Q}{r^2}$$

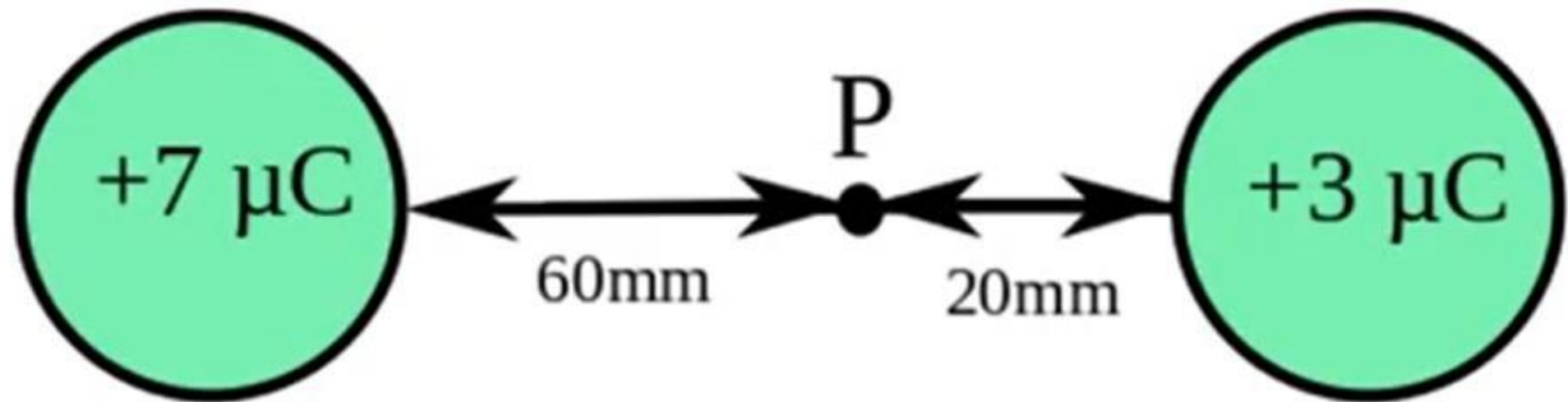
$$E_{new} \propto \frac{5}{(3)^2} E$$

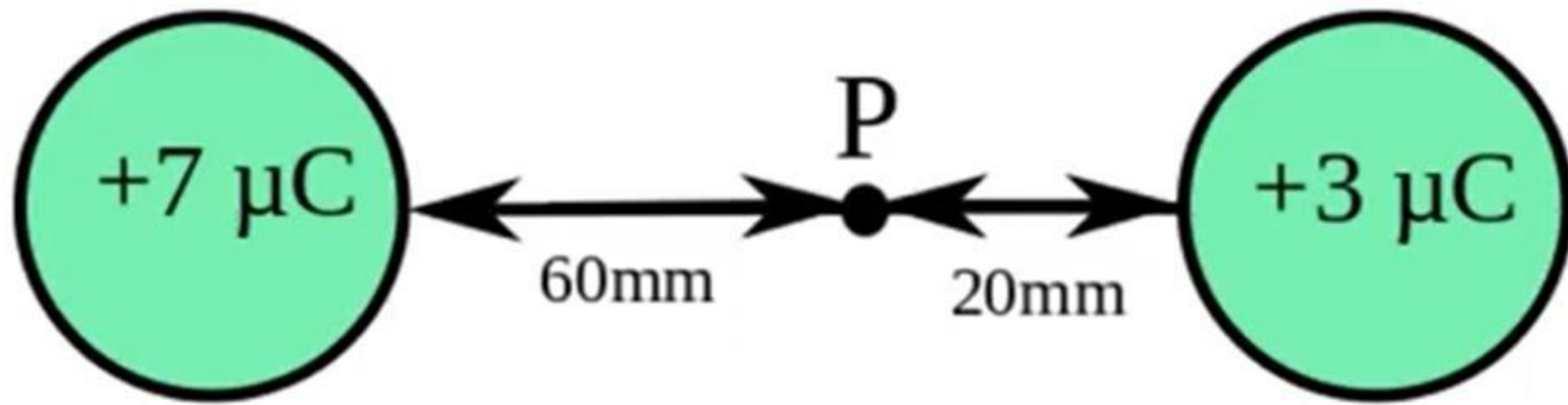
$$E_{new} = \frac{5}{9} E$$

$$E_{new} \propto \frac{5}{9} E$$

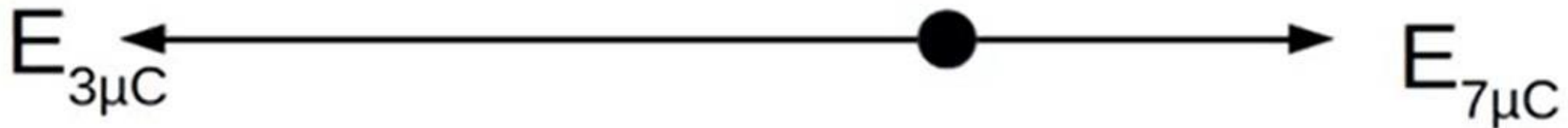
Example

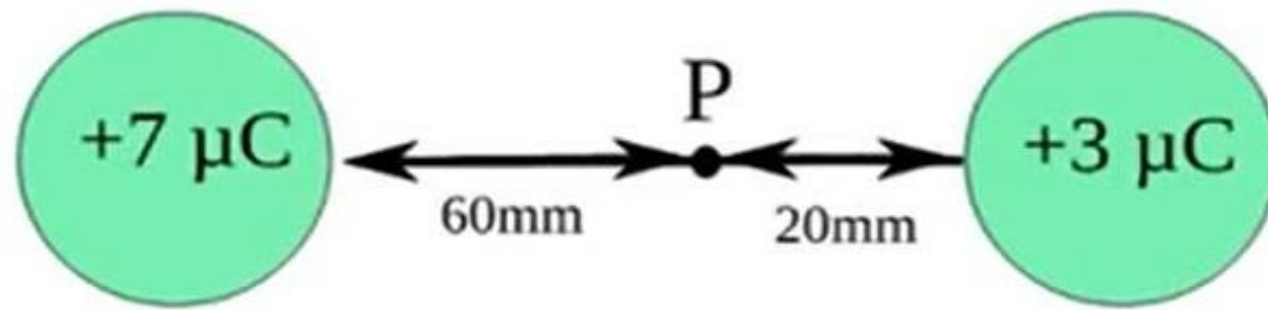
Point P is between two charges of $+7\text{ }\mu\text{C}$ and $+3\text{ }\mu\text{C}$ as shown in the sketch. Calculate the resultant **electric field** at point P. Choose left as positive.



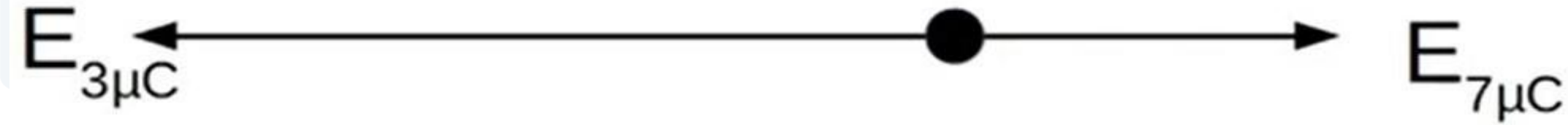


Because these are vectors, they can also be illustrated using a free body/force diagram





Remember :
Choose left as
positive



$$E_{ave+3\mu C} = \frac{kQ}{r^2}$$

$$E_{ave+3\mu C} = \frac{(9 \times 10^9)(3 \times 10^{-6})}{(20 \times 10^{-3})^2}$$

$$E_{ave+3\mu C} = 6,75 \times 10^7 \text{ N} \cdot \text{C}^{-1}$$

to the left

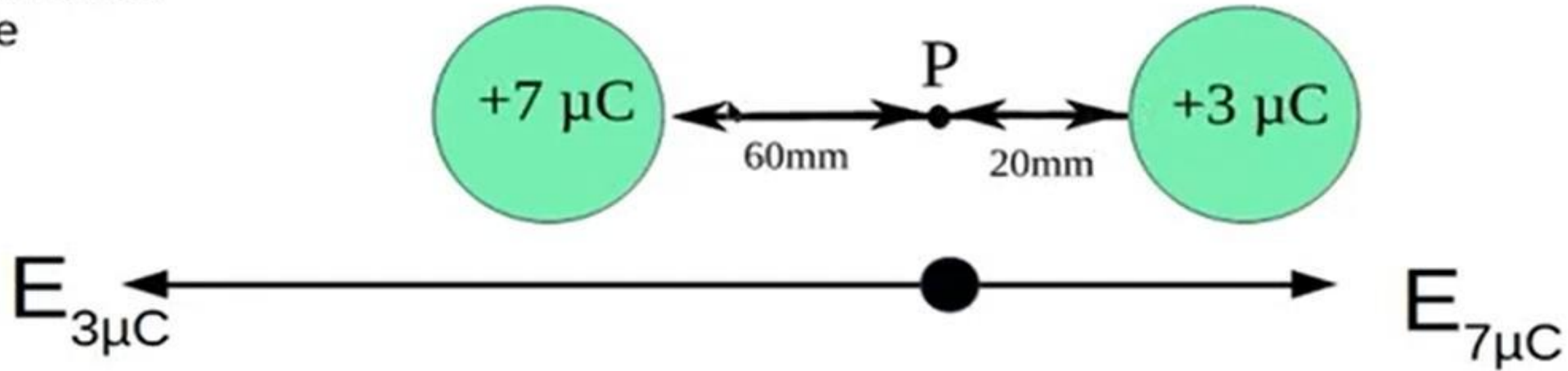
$$E_{ave+7\mu C} = \frac{kQ}{r^2}$$

$$E_{ave+7\mu C} = \frac{(9 \times 10^9)(7 \times 10^{-6})}{(60 \times 10^{-3})^2}$$

$$E_{ave+7\mu C} = 1,75 \times 10^7 \text{ N} \cdot \text{C}^{-1}$$

to the right

Remember :
Choose left as
positive



$$E_{\text{net}} = -E_{7\mu\text{C}} + E_{3\mu\text{C}}$$

$$E_{\text{net}} = -1,75 \times 10^7 + 6,75 \times 10^7$$

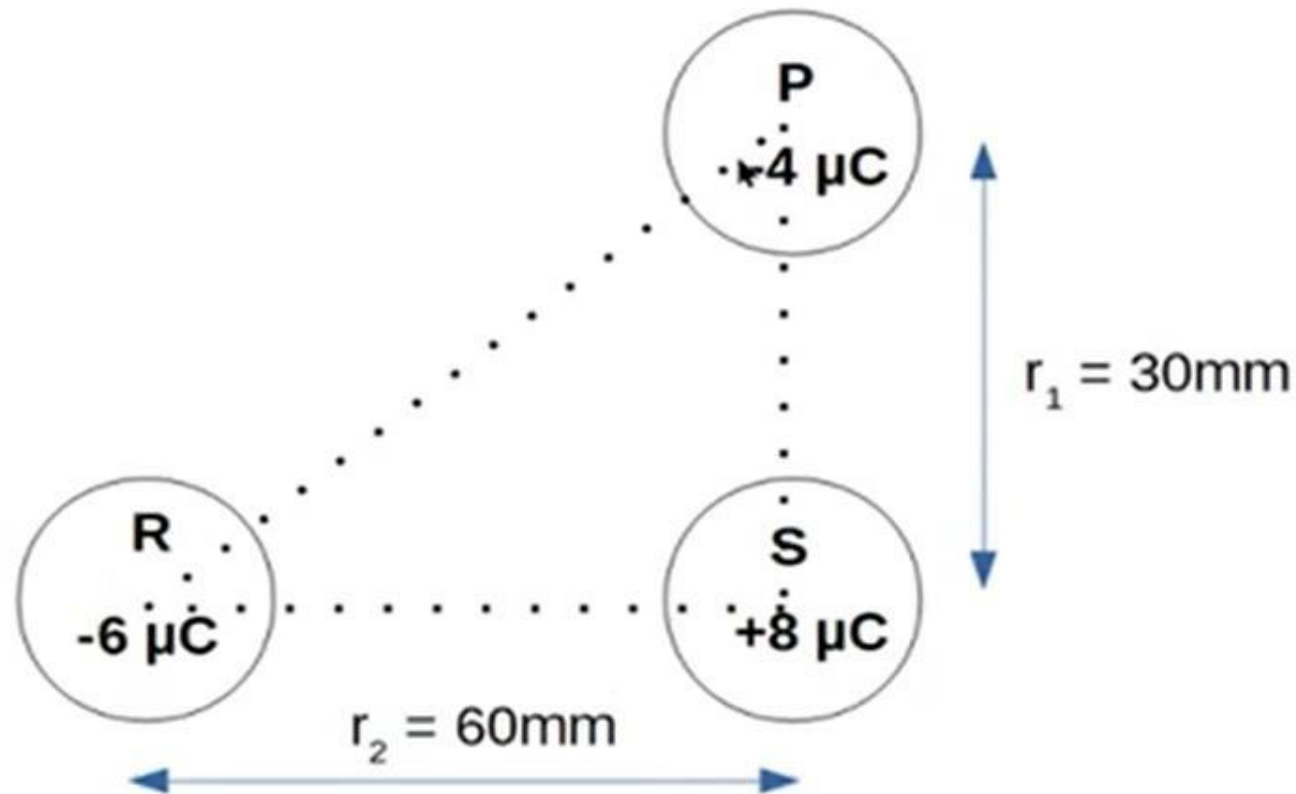
$$E_{\text{net}} = 5 \times 10^7 \text{ N}\cdot\text{C}^{-1} \text{ to the left}$$

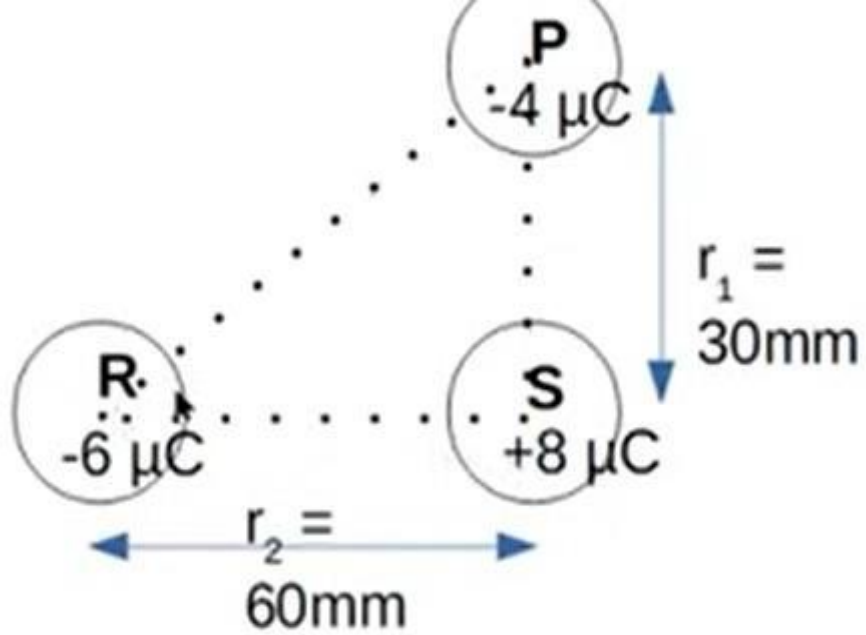
Remember that F and E are
both **vectors** and have
magnitude and **direction**

In two dimensions – (only right angled triangles)

We can calculate the force that P exerts on S, and well as the force of R on S

The forces can then be drawn as **vectors** in a force diagram, (head to tail) and then use **Pythagoras** and **trigonometry** to calculate the direction and magnitude of the **net force**





$$F_{P \text{ on } S} = \frac{k Q_P Q_S}{r_1^2}$$

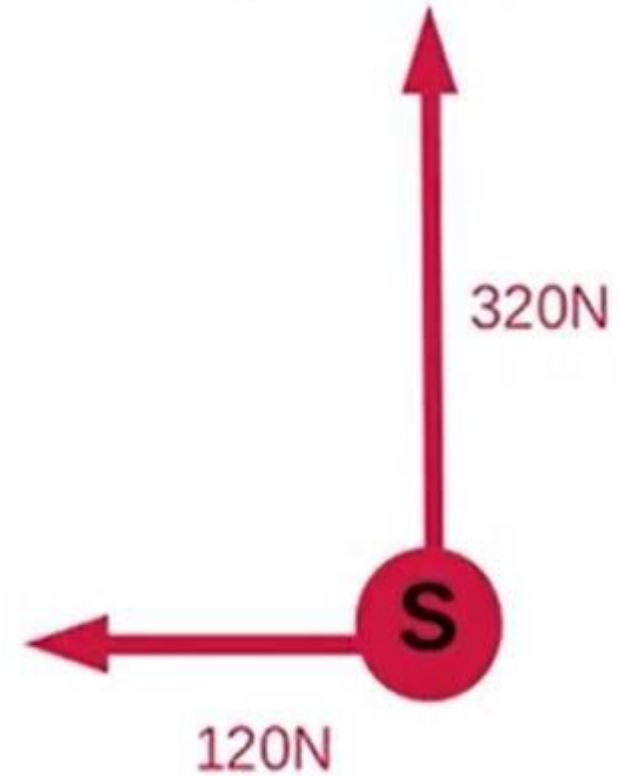
$$F = \frac{9 \times 10^9 (4 \times 10^{-6}) (8 \times 10^{-6})}{(30 \times 10^{-3})^2}$$

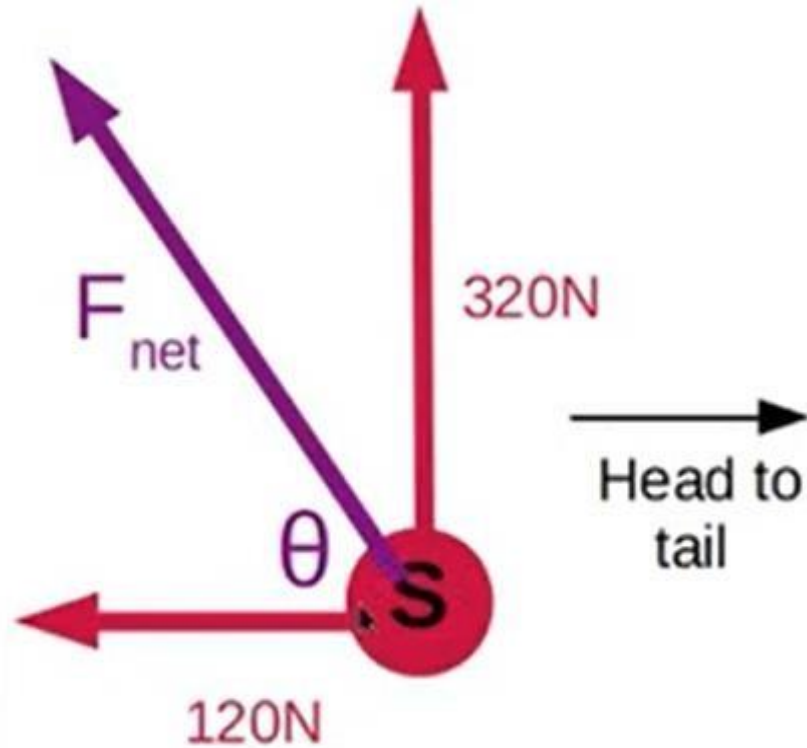
$F = 320 \text{ N}$ attraction upwards to P

$$F_{R \text{ on } S} = \frac{k Q_R Q_S}{r_2^2}$$

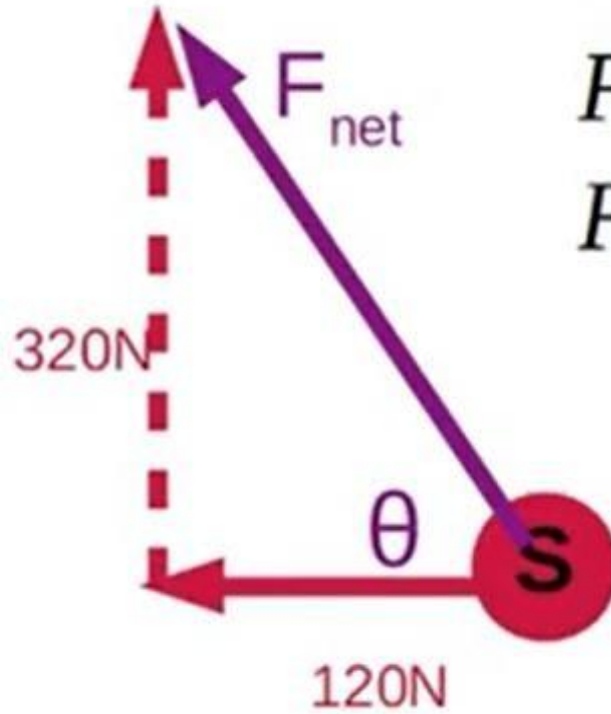
$$F = \frac{9 \times 10^9 (6 \times 10^{-6}) (8 \times 10^{-6})}{(60 \times 10^{-3})^2}$$

$F = 120 \text{ N}$ attraction to the left to R





Head to
tail



$$F_{net\ on\ S} = \sqrt{(F_{P\ on\ S})^2 + (F_{R\ on\ S}^2)}$$

$$F_{net\ on\ S} = \sqrt{320^2 + 120^2}$$

$$F_{net\ on\ S} = 341,76\text{ N}$$

$$\tan \theta = \frac{F_{P\ on\ S}}{F_{R\ on\ S}} = \frac{320}{120}$$

$$\tan \theta = 2,67$$

$$\theta = 69,44^\circ$$

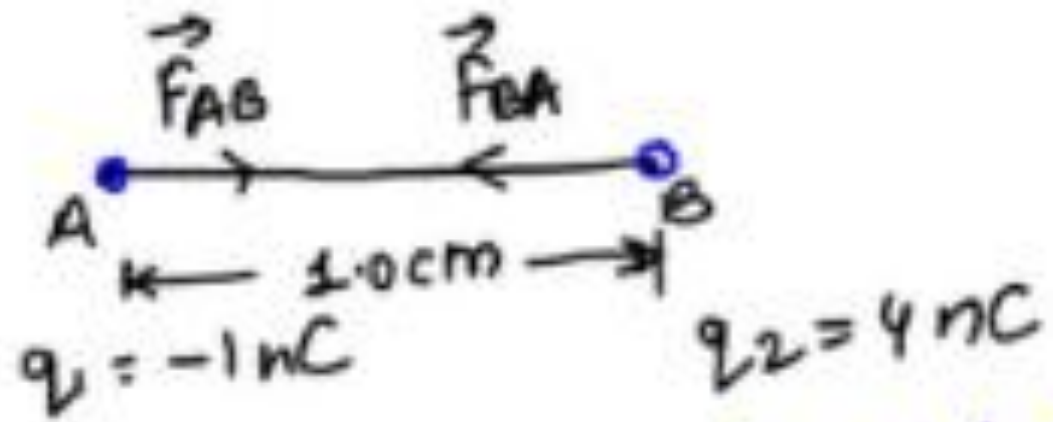
$$F_{net\ on\ s} = 341,76\text{ N at } 69,44^\circ \text{ to } F_{R\ on\ S}$$

What force would a $1.4 \times 10^{-5} \text{C}$ negative charge experience at 1m distance from the 1C positive charge?

Would the force be directed toward or away from the 1C charge?

Point charge A has a charge of -1.0 nC and point charge B has a charge of 4.0 nC .

They are separated by 1.0 cm . What are the magnitude and direction of the electric forces on charges A and B?



$$|\vec{F}| = \frac{1}{4\pi\epsilon_0} \frac{|q_1||q_2|}{r^2}$$

$$= (9 \times 10^9 \text{ Nm}^2/\text{C}^2) (1 \times 10^{-9} \text{ C}) (4 \times 10^{-9} \text{ C})$$

01) The S.I. unit of electric field is

(1) NC^{-1}

(2) Vm^{-2}

(3) Cm

(4) Vm

01) The S.I. unit of electric field is

✓ (1) NC^{-1}

(2) Vm^{-2}

(3) Cm

(4) Vm

$$E = \frac{F}{q} \quad \text{N C}^{-1}$$

$$E = -\frac{dV}{dx} \quad \text{volt m}^{-1}$$

02) The force on an electron of charge q placed at a point in an electric field where the electric intensity is E , is given by $F =$

- (1) E/q in the direction of the field
- (2) qE in the direction of the field
- (3) qE opposite to the direction of the field
- (4) E/q opposite to the direction of the field.

02) The force on an electron of charge q placed at a point in an electric field where the electric intensity is E , is given by $F =$

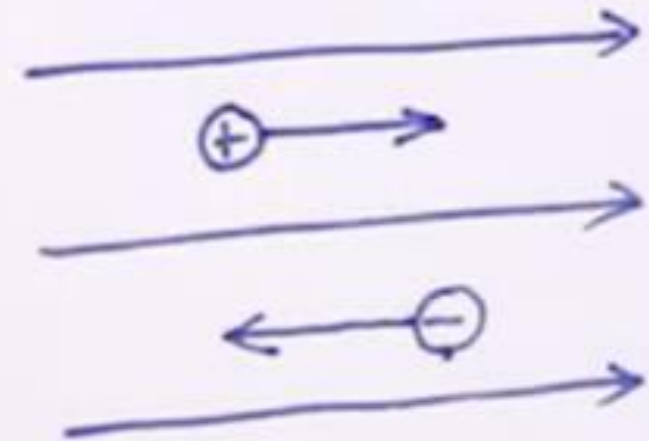
(1) E/q in the direction of the field

(2) qE in the direction of the field ✓

✓ (3) qE opposite to the direction of the field ✓

(4) E/q opposite to the direction of the field.

$$F = qE$$



03) If F be the electrostatic force between two charges in air, then the force between the charges when air is replaced by a medium of dielectric constant K is

(1) $F/\sqrt{K}$

(2) F/K

(3) $\sqrt{K}F$

(4) KF

03) If F be the electrostatic force between two charges in air, then the force between the charges when air is replaced by a medium of dielectric constant K is

(1) $F/\sqrt{K}$

✓ ~~(2) F/K~~

(3) $\sqrt{K}F$

(4) KF

$$\epsilon_r, k = \frac{F_{air}}{F_m}$$

$$F_m = \frac{F}{K}$$

- 04) The magnitude of the electric intensity represented by $\vec{E} = 3\hat{i} + 4\hat{j} - 2\hat{k}$, (where $\hat{i}$, $\hat{j}$ and $\hat{k}$ are unit vectors along x, y and z axes respectively) is
- (1) 5 Vm^{-1} (2) 5.4 Vm^{-1}
(3) 9 Vm^{-1} (4) 81 Vm^{-1}

- 04) The magnitude of the electric intensity represented by $\vec{E} = 3\hat{i} + 4\hat{j} - 2\hat{k}$, (where $\hat{i}$, $\hat{j}$ and $\hat{k}$ are unit vectors along x, y and z axes respectively) is
- (1) 5 Vm^{-1} ☒ (2) 5.4 Vm^{-1}
(3) 9 Vm^{-1} (4) 81 Vm^{-1}

$$E = \sqrt{3^2 + 4^2 + (-2)^2}$$
$$= \sqrt{9 + 16 + 4}$$
$$= \sqrt{29}$$
$$\sqrt{25}$$

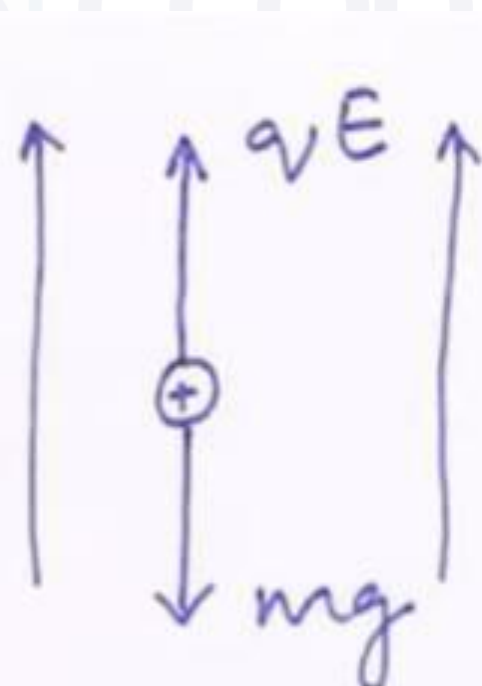
- 06) The magnitude of the electric intensity represented by $\vec{E} = 3\hat{i} + 4\hat{j}$, (where $\hat{i}$ and $\hat{j}$ are unit vectors along X and Y axes respectively), then the Z component of electric field is
- | | |
|-------------------------|-------------------------|
| (1) 5 Vm^{-1} | (2) 7 Vm^{-1} |
| (3) 1 Vm^{-1} | (4) 0 Vm^{-1} |

The electric field in which a charged oil drop of mass 5 mg and charge $+ 10 \mu\text{C}$ be held stationary is

- (1) 5000 Vm^{-1} upwards
- (2) 5 Vm^{-1} downwards
- (3) 5 Vm^{-1} upwards
- (4) 5000 Vm^{-1} downwards

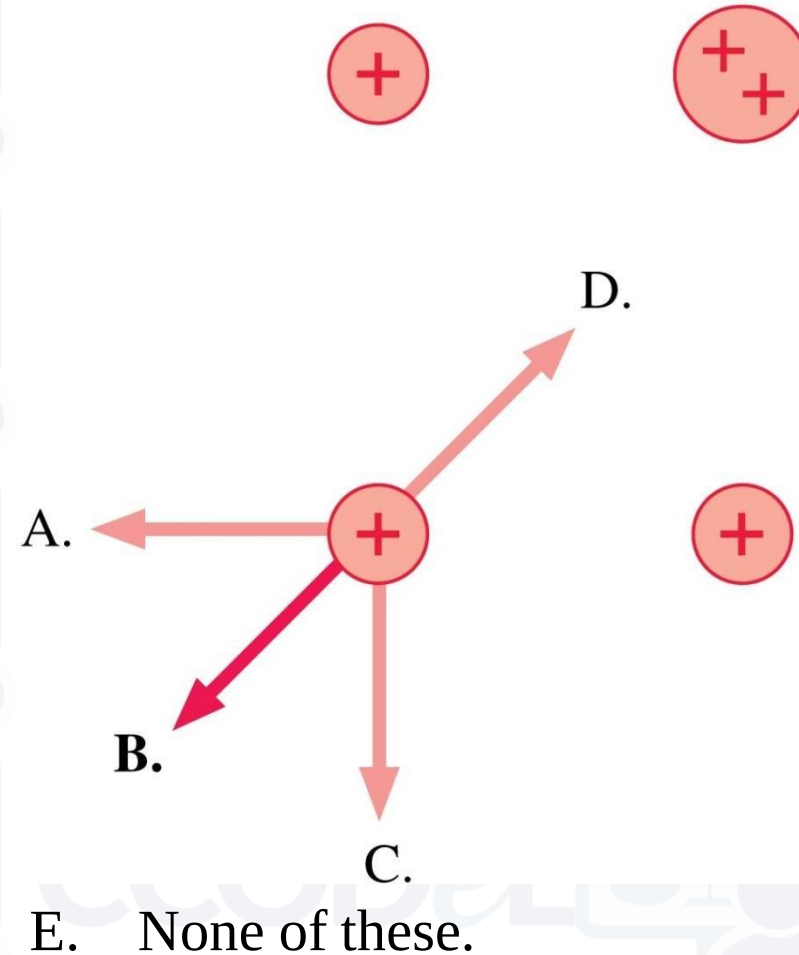
The electric field in which a charged oil drop of mass 5 mg and charge + 10 μC be held stationary is

- (1) 5000 Vm^{-1} upwards ✓
- (2) 5 Vm^{-1} downwards ✗
- (3) 5 Vm^{-1} upwards ✓
- (4) 5000 Vm^{-1} downwards



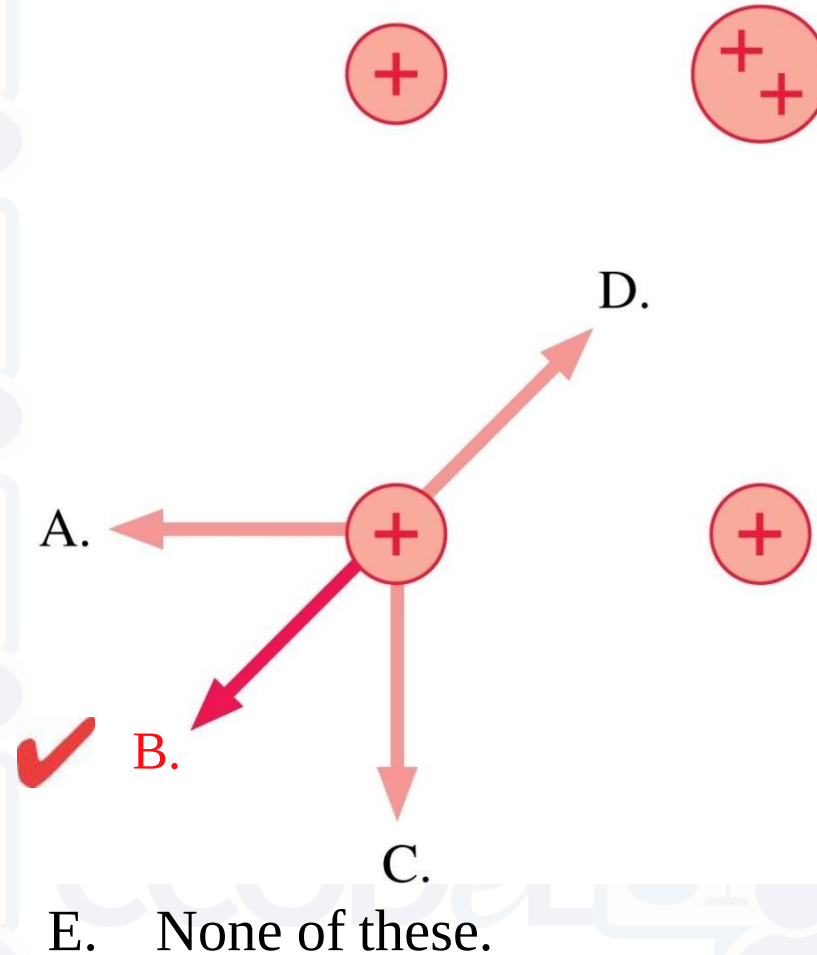
$$\begin{aligned}
 qE &= mg \\
 E &= \frac{(5 \times 10^{-3}) \times 10^{-6}}{10 \times 10^{-6}} \\
 &= 5 \text{ NC}^{-1} \text{ Vm}^{-1}
 \end{aligned}$$

Which is the direction of the net force on the charge at the lower left?

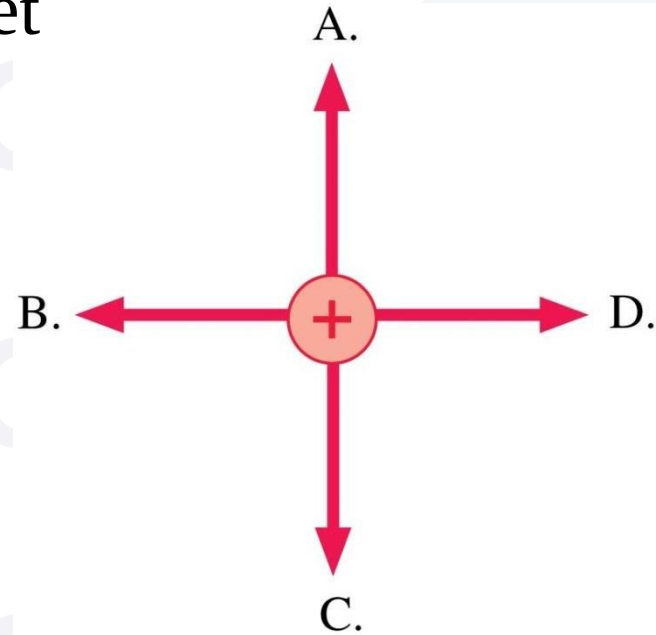


E. None of these.

Which is the direction of the net force on the charge at the lower left?



Which is the direction of the net force on the charge at the top?

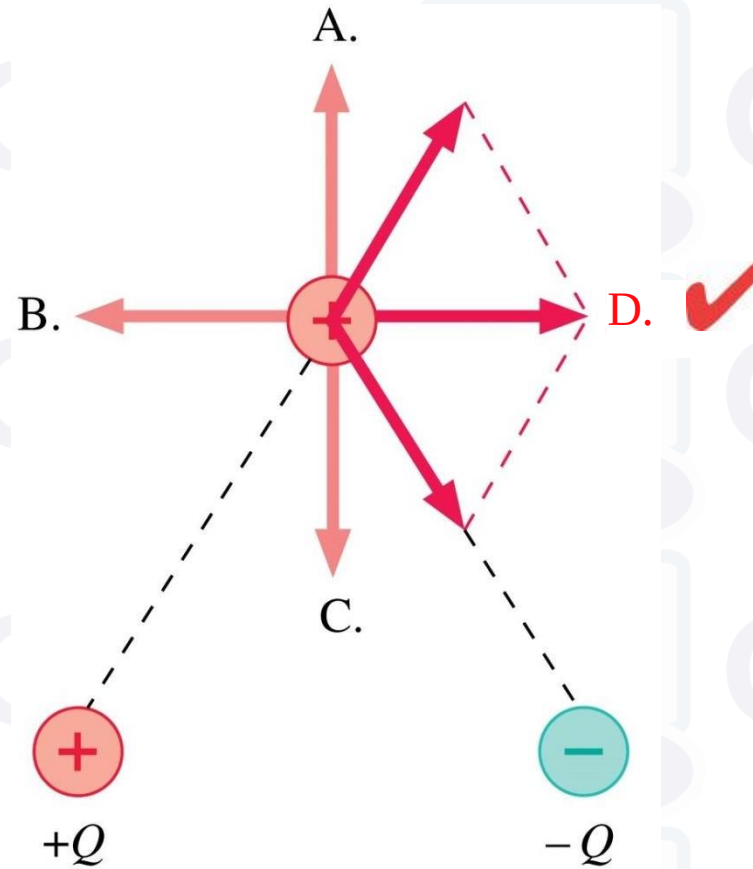



 $+Q$


 $-Q$

E. None of these.

Which is the direction of the net force on the charge at the top?



E. None of these.

CLASS WORK

- 1. What is the direction of the field 1 m away from the 1C charge, in or out of the 1C charge?**
- 2. What force would a $1.4 \times 10^{-5} \text{C}$ negative charge experience at 1m distance from the 1C positive charge?**
- 3. Would the force be directed toward or away from the 1C charge?**
- 4. Point charge A has a charge of -1.0 nC and point charge B has a charge of 4.0 nC . They are separated by 1.0 cm . What are the magnitude and direction of the electric forces on charges A and B?**



MODULE 1: ELECTROSTATICS

UNIT 3: Electric potential and energy

INTRODUCTION

- In electrostatics, electric charges exert forces on one another. When charges interact, work is done, and energy is either stored or transferred.
- The concepts of electric potential and electric potential energy help explain how electrical energy behaves in electric fields.
- Electric potential is analogous to gravitational potential in mechanics.
- Just as an object at a height possesses gravitational potential energy, a charge in an electric field possesses electric potential energy.

LEARNING OUTCOMES

At the end of this unit, you should be able to :

- **Compute electric potential due to point charges**
- **Relate the electric field to the potential gradient**
- **Calculate work done in moving charges in electric fields**

- **Electric potential energy is the energy possessed by a charged particle due to its position in an electric field.**
- **It is the work done in bringing a charge from infinity to a point in an electric field without acceleration.**

For two point charges:

$$U = k \frac{q_1 q_2}{r}$$

Where:

- U = electric potential energy
- k = Coulomb constant
- q_1, q_2 = charges
- r = separation distance

When a charge moves in an electric field, work is done.

The work done is given by:

$$W = qV$$

Where:

- W = work done
- q = charge
- V = potential difference

Electric Potential

- Electric potential at a point is the work done per unit positive charge in bringing the charge from infinity to that point.

$$V = \frac{W}{q}$$

Where:

- V = electric potential
- W = work done
- q = charge

The electric potential produced by a point charge is:

$$V = k \frac{Q}{r}$$

Characteristics of Electric Potential

- It is a scalar quantity.
- It can be positive or negative.
- It depends on the source charge and distance.
- It obeys the superposition principle.

Electric Potential Difference

- Potential difference between two points is the work done per unit charge in moving a charge from one point to another.

$$\Delta V = \frac{W}{q}$$

Relationship Between Electric Field and Potential

Electric field intensity is related to the rate of change of electric potential.

$$E = -\frac{dV}{dr}$$

Where:

- E = electric field intensity
- V = electric potential
- r = distance

The negative sign indicates that electric potential decreases in the direction of the electric field.

Questions

1. Calculate the electric potential at a point 2 m from a charge of $5 \times 10^{-6} \text{ C}$.
2. A charge of 3C moves through a potential difference of 12V. Calculate the work done.
3. Explain why no work is done along an equipotential surface.

Equipotential Surfaces

- An equipotential surface is a surface where all points have the same electric potential.
- Properties of Equipotential surfaces
 1. No work is done when moving a charge along an equipotential surface.
 2. Electric field lines are perpendicular to equipotential surfaces.
 3. Equipotential surfaces never intersect.



MODULE 1: ELECTROSTATICS

UNIT 4: gauss's law

INTRODUCTION

- **Gauss's Law is one of the fundamental laws of electrostatics in Electromagnetism.**
- **It relates the electric field around a charged object to the total charge enclosed within a surface**
- **It provides a powerful method for calculating electric fields, especially when there is symmetry (spherical, cylindrical, or planar).**

Before understanding Gauss's Law, recall that:

- An electric field (E) is the force experienced per unit positive charge.
- It is given by:

$$E = \frac{F}{q}$$

- Unit: Newton per Coulomb (N/C)

Electric flux measures how much electric field passes through a surface.

Definition

Electric flux (Φ) is given by:

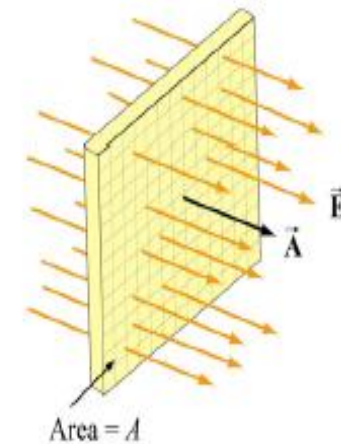
$$\Phi = EA \cos \theta$$

Where:

- E = electric field
- A = area
- θ = angle between field and normal to the surface

Key Idea

- More field lines crossing a surface $\rightarrow$ higher flux
- No field lines $\rightarrow$ zero flux



Electric field lines passing through a surface of area A .

Gauss's Law states:

“The total electric flux through a closed surface is proportional to the total charge enclosed within that surface.”

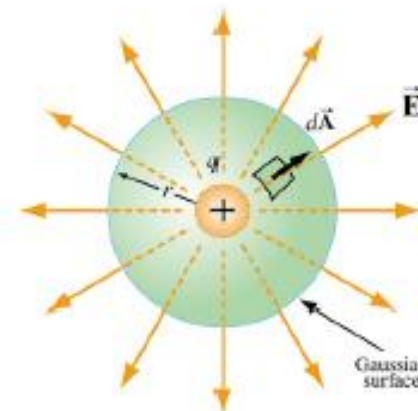
Mathematical Form

$$\oint \mathbf{E} \cdot d\mathbf{A} = \frac{Q_{\text{enc}}}{\epsilon_0}$$

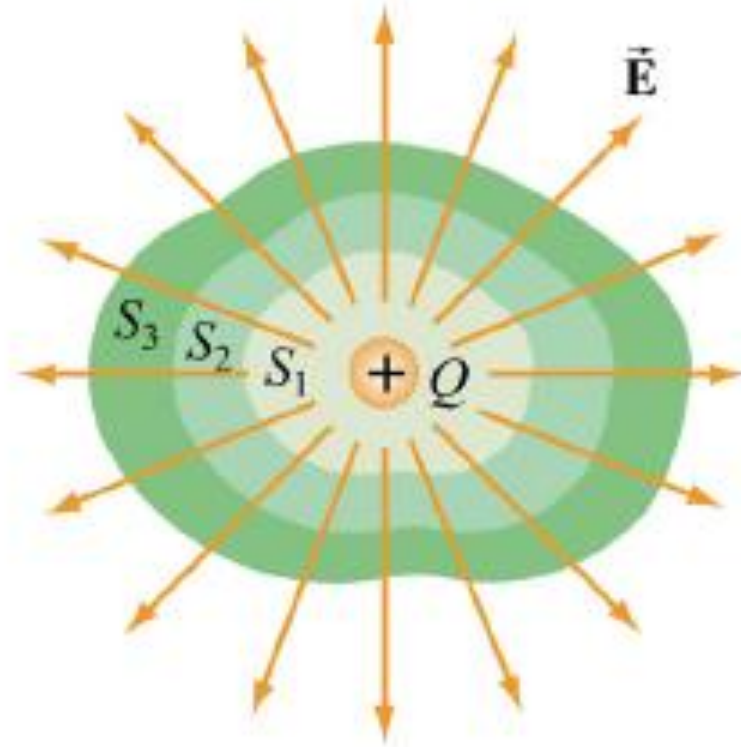
$$\oint \mathbf{E} \cdot d\mathbf{A} = \frac{Q_{\text{enc}}}{\epsilon_0}$$

Where:

- $\oint$ = integral over a closed surface
- Q_{enc} = enclosed charge
- ϵ_0 = permittivity of free space



A spherical Gaussian surface enclosing a charge Q .



Different Gaussian surfaces with the same outward electric flux.

Figure above, the same result ($\Phi_E = Q / \epsilon_0$) is obtained. whether the choice is S_1 , S_2 or S_3 .

Gaussian Surfaces

A Gaussian surface is an imaginary closed surface used to apply Gauss's Law.

Examples

- Sphere
- Cylinder
- Cube

Important Note

Only the charge inside the surface contributes to flux.

Physical Interpretation

- **Electric field lines originate from positive charges and terminate on negative charges**
- **The number of lines passing through a surface depends on enclosed charge**
- **External charges do NOT affect net flux through the surface**

Case 1: Point Charge

- Use Gauss's Law to find the electric field at a distance r from a point charge Q .
- Choose spherical Gaussian surface
- Area = $4\pi r^2$
- Apply Gauss's Law:

$$E(4\pi r^2) = \frac{Q}{\epsilon_0}$$

$$E = \frac{Q}{4\pi\epsilon_0 r^2}$$

Case 2: Infinite Line Charge

- Find the electric field at distance r from a long straight wire with charge density λ

Use a cylindrical Gaussian surface:

- Surface area = $2\pi rL$
- Charge enclosed = λL

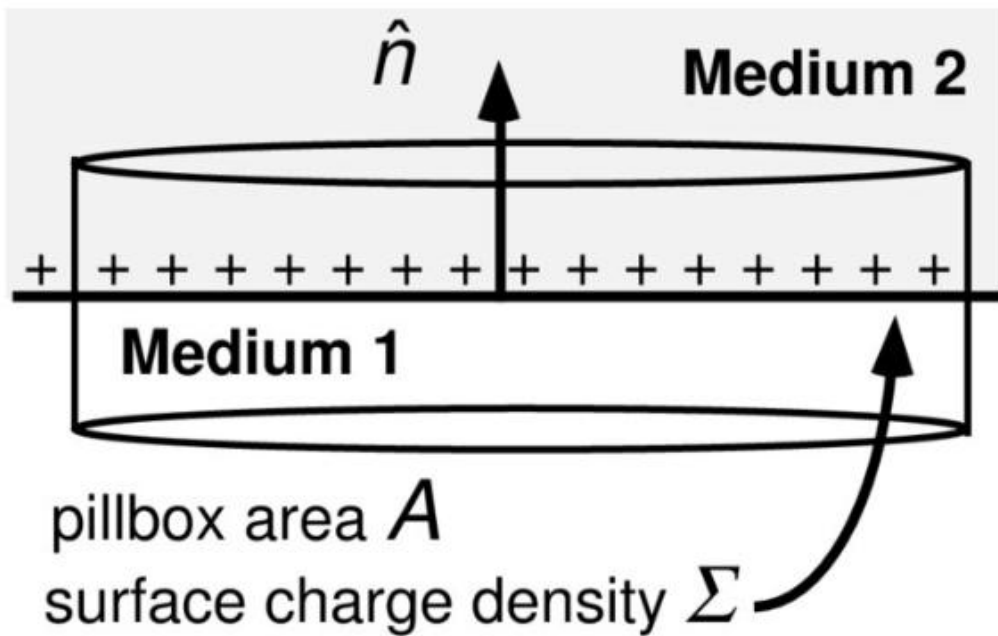
$$E(2\pi rL) = \frac{\lambda L}{\epsilon_0}$$

$$E = \frac{\lambda}{2\pi\epsilon_0 r} \quad E \propto \frac{1}{r}$$

Case 3: Infinite Plane Sheet

- Find the electric field near an infinite plane sheet with charge density σ

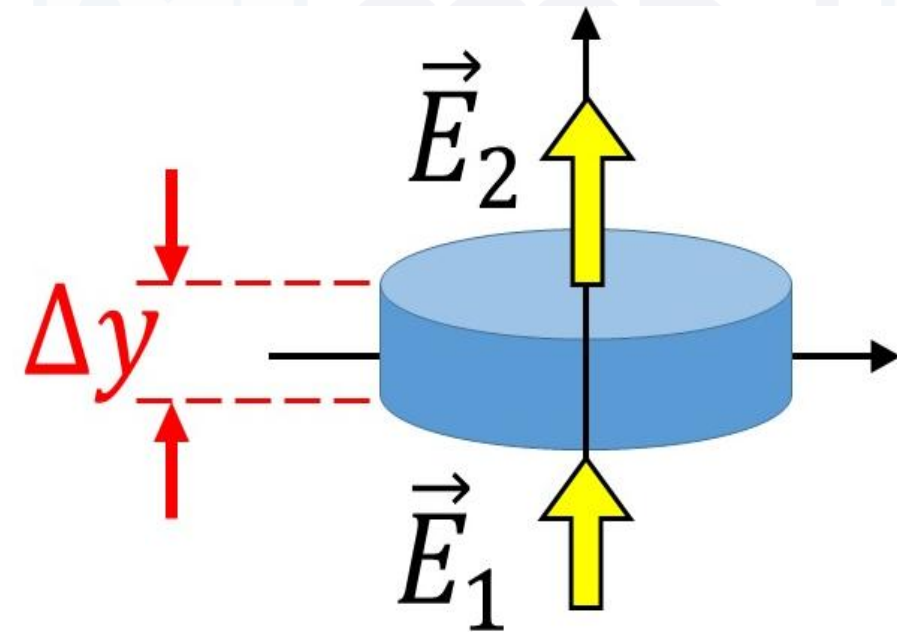
Use a pillbox Gaussian surface:



$$2EA = \frac{\sigma A}{\epsilon_0}$$

$$E = \frac{\sigma}{2\epsilon_0}$$

- Field is constant
- Independent of distance



Case 4: Electric Flux Through a Surface

A uniform electric field $E = 200 \text{ N/C}$ passes through a surface of area 2 m^2 at 0° . Find flux.

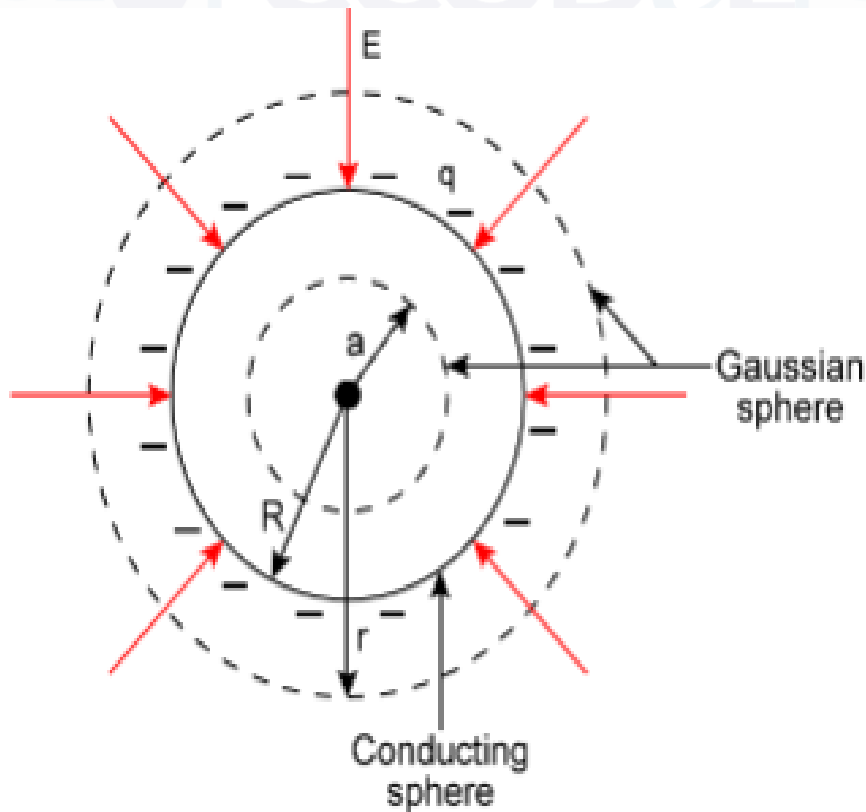
$$\Phi = EA \cos \theta$$

$$\Phi = 200 \times 2 \times \cos 0^\circ$$

$$\Phi = 400 \text{ Nm}^2/\text{C}$$

Case 5: Electric Field Inside a Hollow Sphere

- Find the electric field inside a hollow conducting sphere.



Outside the conducting surface:

The radius of the Gaussian surface drawn outside the conducting sphere is 'r' and that for conducting sphere is R. Hence $r > R$

Let the charge enclosed by this Gaussian surface be q
So the total electric field over the Gaussian surface ($r > R$) is,

$$E = \frac{-q}{\epsilon_0 A}$$

$$A = 4\pi r^2$$

$$\therefore E = \frac{-q}{\epsilon_0 4\pi r^2}$$

• Inside the conducting surface:

- Now, similarly, we will draw the Gaussian surface inside the conducting surface.
- In the given figure, you can see the Gaussian surface inside the conducting sphere of radius 'a'
- This Gaussian surface encloses no charge.
- So using Gauss theorem, $E=0$
- So as all the charges lie on the surface of conducting sphere, using symmetry and Gauss law the electric field is zero inside the hollow conducting sphere.
- So, we can say that the electric field inside any hollow conducting surface is zero provided that there are no charges enclosed by that region.

- Choose Gaussian surface inside the sphere
- Enclosed charge = 0

$$E = 0$$



CLASS WORK

1. A point charge of $5 \times 10^{-6} \text{ C}$ is placed at the center of a spherical Gaussian surface of radius 0.20 m. Calculate the electric field at the surface of the sphere.
2. A closed spherical surface encloses a charge of $3 \mu\text{C}$. Calculate the total electric flux through the surface.
3. An infinitely long straight wire carries a uniform linear charge density of $4 \times 10^{-6} \text{ C/m}$. Using a cylindrical Gaussian surface of radius 0.05 m, calculate the electric field at that distance from the wire.
4. A solid sphere of radius 0.10 m has a uniform charge density of $2 \times 10^{-6} \text{ C/m}^3$. Using Gauss's law, determine the electric field at a distance of 0.05 m from the center.
5. A conducting sphere of radius 0.25 m carries a total charge of $6 \times 10^{-6} \text{ C}$. Calculate the electric field at a point 0.50 m from the center of the sphere.